

Computer Science XII

Exam revision cards — definitions, tables, mnemonics and ★
Golden topics on one glance each.

Bilingual support: English + اردو

امتحانی چیٹ شیٹ — تعریفیں، جدول، یادداشت اور گولڈن موضوعات

01 Computer Systems (HCI)

02 Computational Thinking & Algorithms

03 Programming Fundamentals

04 Data and Analysis

05 Application and Impacts of Computing

06 Entrepreneurship in the Digital Age

XII

Contents

32 Golden cards are listed below. Start there if you have less than two hours.

Chapter 1: Computer Systems (HCI) — 16 cards

Chapter 2: Computational Thinking & Algorithms — 24 cards

Chapter 3: Programming Fundamentals — 18 cards

Chapter 4: Data and Analysis — 9 cards

Chapter 5: Application and Impacts of Computing — 17 cards

Chapter 6: Entrepreneurship in the Digital Age — 18 cards

Night-before recaps: one-page summaries of every chapter

★ Golden index — if you only have two hours

These are the high-yield topics. Say each definition aloud, then draw the table or diagram from memory.

صرف دو گھنٹے ہوں تو یہ گولڈن کارڈز زبانی دہرائیں، پھر جدول حافظے سے بنائیں۔

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How to use these Cheat Sheets

This booklet is the **night-before and last-week** revision pack. Each topic is one card: the definition you must write, the table or formula, a memory trick, the usual exam wording, and one Urdu line. It is not a textbook. Open the Student's Edition when you need a worked example.

یہ چیٹ شیٹ امتحان سے پہلے رات اور آخری ہفتے کے لیے ہے۔ ہر موضوع ایک کارڈ ہے۔ مثالیں اصل کتاب میں ہیں۔

How to revise with this booklet

1. **Two hours left.** Read only the ★ Golden cards and the Golden index. Say each definition aloud.
2. **One day left.** One chapter at a sitting. Cover the card, write the table from memory, then check.
3. **The night before.** Read the night-before recaps. Do not start a new chapter.
4. **In the hall.** Remember the recipe: *Define. Explain. Example. Diagram. Working.*

What each card holds

- **Definition** — the 1-mark sentence. Write it exactly.
- **Points / table / snippet** — the 3-mark and 5-mark body.
- **Remember** — a mnemonic or trap to avoid.
- **Exam** — how the board asks it.
- **اردو** — the same idea in one spoken line.

★ **Golden** cards are high-yield. If time is short, skip non-Golden cards first, not these.

CHAPTER 1

Computer Systems (HCI)

★ 5 Golden · 16 cards · Applications of HCI (Domains of Life) · Importance of HCI · Human-Computer Interaction Problems · User Interface Design (UI vs UX) · Testing Methods (Usability, A/B, Automated)

1.1 Fundamentals of Human-Computer Interaction

Human-Computer Interaction (HCI) is the study and design of technology so that it serves people.

- An HCI system is easy to use, efficient, and responsive to human needs.
- HCI is also known as CHI (Computer-Human Interface) or MMI (Man-Machine Interaction).
- The three names mean the same idea: the meeting point of a person and a machine.

Channel	How humans use it	How computers respond
Sight (بصارت)	Look at screens, graphics or videos.	Displays visuals on monitors and screens.
Touch (چھونا)	Tap, type or swipe.	Responds with clicks or vibrations (haptics).
Hearing (سننا)	Listen to sounds or voices.	Provides audio feedback via speakers.
Voice (آواز)	Speak commands (for example, "Alexa").	Processes speech and replies.
Spatial (مکانی)	Move in physical space.	Tracks position or body movement (for example, VR).

Remember: Remember the three names: HCI = CHI = MMI. Think "Human meets Machine." Five channels: S-T-H-V-S — Sight, Touch, Hearing, Voice, Spatial. "See The Handy Voice..."

بیومن کمپیوٹر انٹرایکشن (HCI) کا مطلب ہے ٹیکنالوجی کو اس طرح ڈیزائن کرنا کہ وہ انسانوں کے لیے فائدہ مند، آسان اور مؤثر ہو۔

1.1.1 Traditional Interaction and 1.1.2 Natural Interaction

1.1.1 Traditional interaction is the basic way people use everyday devices.

- Computer: moving a mouse and typing on a keyboard; output via monitor and speakers.
- Smartphone (traditional use): tapping and swiping on a touchscreen; output via screen lighting and vibrations.
- Voice: speaking commands into a device (for example, "Alexa, play a song").
- Body and gesture: facial recognition to unlock a phone, or playing VR games by waving hands.

Feature	Traditional	Natural
Main input	Mouse, keyboard, keypad, simple tap/swipe	Voice, face, hand/body gesture, movement
Feels like	Operating a machine	Talking or moving as you do with people
Examples	TV remote, ATM keypad, school-portal login typed on a PC	FaceID, Siri, Kinect games, "Hey Google"
Typical output	Screen, speakers, vibration	Spoken reply, unlock, tracking of the body in VR

Remember: Test sentence: If you must press keys or tap labelled buttons, it is traditional. If the machine understands speech, face or body movement, it is natural.

روایتی تعامل (Traditional Interaction) میں ماؤس، کی بورڈ، کی پیڈ یا عام ٹیپ اور سوائپ استعمال ہوتے ہیں، جیسے اے ٹی ایم کی پیڈ۔

1.2 Applications of HCI (Domains of Life) ★ Golden

HCI is used to design and develop interactive systems such as desktop applications, websites and mobile apps.

- These are human-centred solutions built to meet user needs across many domains of life.
- A domain is an area of daily activity — health, money, school, talking to people.
- When you design for a domain, you design for that domain's users, tasks and risks.

Domain	Application examples	User benefit
Health care	Patient portals, telemedicine apps, assistive technologies	Improves access to medical services remotely
Banking	Mobile banking apps, secure ATM interfaces	Enables quick transactions and secure financial management
Education	E-learning platforms (LMS), online exam portals	Enhances teaching, learning and student engagement
Networking and communication	Social media, messaging apps (WhatsApp), Zoom	Connects people globally through collaboration tools

Remember: Exam mnemonic H-BEN: Health, Banking, Education, Networking. For each, write: one application + one user benefit. That structure scores full marks on...

Exam: Very common 5–8 mark question: "Explain applications of HCI in different domains of life with examples." Draw a four-row table (Health care, Banking, Education,...)

ایچ سی آئی کی ایپلیکیشنز صحت، بینکنگ، تعلیم اور نیٹ ورکنگ کے شعبوں میں انسانی ضروریات پوری کرتی ہیں۔

1.3 Components of HCI

HCI relies on several primary components that work together.

- If one component is weak, the whole interaction feels broken.
- The first four are the people, the machine, the conversation, and the job to be done.

	Component	Meaning
i	User (صارف)	Any individual or group that interacts with a computer system to achieve defined goals. Two categories: end users (people who directly operate the system, from beginners to experts) and stakeholders (people with an interest in the outcome, such as managers and clients).
ii	Computer system (کمپیوٹر سسٹم)	The platform where humans and computers communicate through input, output and feedback using an interface. It connects human actions with machine responses.
iii	User interaction (صارف کا تعامل)	The process through which a user communicates with a system by performing actions (clicking, tapping, typing, speaking). It transforms user intent into system responses.
iv	Task (کام)	The specific action users perform (for example, booking a ticket). It has three characteristics: task complexity (filling a form vs designing a 3D model), user knowledge (basic typing vs advanced programming), and time requirement (ATM withdrawal vs online passport application).

Remember: U-C-I-T: User, Computer system, Interaction, Task. "U Can Interact with Tasks." Then add three task questions: How hard? What skill? How long?

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد ہے)، کمپیوٹر سسٹم (ا...)

Types of User Interaction

User interaction can happen in many different ways depending on the technology used.

- Learn every type with one clean example — board questions often ask you to "list and explain" five or more.

Remember: Group them: hands on screen (direct, gesture, haptic) · words (command, conversational, voice) · lists and blanks (menu, form) · automatic/body (context-aware,...

تعامل کی اقسام میں براہ راست ہیرا پھیری، کمانڈ، مینو، فارم، گفتگو (چیٹ بوٹ)، اشارہ، آواز، سیاق و سباق سے آگاہ نظام، بائیومیٹرک، بی...

Interface Types, Environment and Feedback

Components v-vii complete the HCI picture: the look of the controls, the place where you use them, and the signals that tell you "it worked" or "try again".

- Graphical User Interface (GUI): visual elements such as windows, icons and menus (Windows OS, Android).
- Command-Line Interface (CLI): text-based typing (MS-DOS, Linux terminal).
- Touch interface: tapping, swiping, pinching (smartphones, tablets).
- Voice User Interface (VUI): spoken commands (smart speakers).

Remember: GUI vs CLI vs NUI: GUI = pictures you click. CLI = words you type. NUI = body you move. VUI = voice you speak. Touch = fingers on glass. Feedback rule: system...

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، ٹچ، آواز (VUI) اور قدرتی اشارے (NUI).

1.4 Importance of HCI ★ Golden

HCI ensures that machines are approachable and usable, turning complex technology into everyday tools.

- Good design makes technology simple and enjoyable (touchscreens).
- Poor design causes confusion and risks (unclear buttons on medical devices).
- This is why HCI is not optional decoration.

Implication	Description
1. Enhance productivity	Well-designed interfaces help users work faster, make fewer mistakes, and achieve goals efficiently.
2. Promotes inclusivity	Accessibility features (screen readers, voice commands) empower marginalized groups and ensure equal use of technology for everyone.
3. Drives innovation	Emerging tech such as AR/VR, AI and IoT rely strictly on solid HCI principles for people to actually adopt and use them.
4. Economic impact	User-friendly products gain higher adoption rates, customer loyalty and market success, driving profits.
5. Ethical responsibility	Responsible design ensures technology respects user privacy, reduces bias and supports human well-being.

Remember: P-I-I-E-E: Productivity, Inclusivity, Innovation, Economic impact, Ethics. Sentence: "Please Include Innovative, Ethical Earnings."

Exam: Typical wording: "Why is HCI important? Discuss its implications." (8 marks) Write a short opening (good design vs poor design) then five numbered points with one...

اچھا ایچ سی آئی پیچیدہ ٹیکنالوجی کو روزمرہ کا آسان اوزار بناتا ہے، برا ڈیزائن الجھن اور خطرہ پیدا کرتا ہے، جیسے طبی آلے پر غیر واضح بٹن۔

1.5 Accessibility Principles

Accessibility principles are HCI guidelines that ensure digital systems are usable by all individuals, including people with diverse physical, sensory and cognitive abilities.

- They remove barriers to create equal access.
- Accessibility is not a "special extra".
- It is how you make sure a student with low vision, a shopkeeper who cannot use a mouse, or a colour-blind driver can still complete the...

Principle	What it means	Example
1. High contrast (واضح رنگ)	Make text and pictures stand out clearly against the background.	Black text on a white background in a railway app, readable in bright sunlight.
2. Readable fonts (پڑھنے میں آسان تحریر)	Clear, simple fonts with proper size and spacing.	A school website using large Calibri so homework instructions are easy to read.
3. Colour-blind friendly design (رنگوں کی پہچان)	Do not depend <i>only</i> on colours; also use symbols or text labels.	A traffic app showing a red circle with a "STOP" label, not just a red colour.
4. Inclusive navigation (آسان نیویگیشن)	Give users different ways to move around a system.	A website that lets you use the Tab key to jump between buttons if you cannot use a mouse.
5. Multimodal detection (متعدد حسی ذرائع)	Present information in different formats (sight, sound, touch).	A video with text captions, or a phone vibrating to alert a user.

Remember: C-F-C-N-M: Contrast, Fonts, Colour-blind, Navigation, Multimodal. "Can Friends Catch New Messages?"

رسائی پذیری (Accessibility) کے اصول یقینی بناتے ہیں کہ ہر انسان سسٹم استعمال کر سکے، بشمول مختلف جسمانی، حسی اور ذہنی صلاحیتوں والے افراد۔

1.6 Need Analysis for Interface Design

Need analysis is the process of identifying the gap between the existing situation and the desired outcome.

1. Identify the users → 2. Collect information → 3. Analyse needs → 4. Prioritise → 5. Develop solutions

- In HCI it is done before designing a UI, so that you understand users' goals, tasks, abilities and expectations.
- If you skip need analysis, you guess.
- Guessing is how a school portal ends up with 40 menus and no "Download date sheet" button on the home screen.

Remember: Need analysis = gap finding. Write: Existing = what hurts now. Desired = what "done" looks like. The UI must close that gap, not decorate it.

نیڈ اینالیسس موجودہ صورتحال اور مطلوبہ نتیجے کے فرق کو یو آئی بنانے سے پہلے تلاش کرتا ہے۔

1.7 Human-Computer Interaction Problems ★ Golden

HCI problems are common difficulties that stop users from using technology smoothly.

- They happen when systems fail to meet human needs or cognitive limits.
- Cognitive limits means the mind can only hold and process so much at once — if an app asks you to remember three SMS codes at the same...
- Board answers score well when you name the problem, explain it in one line, and give a concrete example (the source examples are safe; a...

Problem	Explanation and example
Interface complexity	Too many options or a confusing layout. Example: a shopping app with so many menus you cannot find the checkout button.
Poor feedback	The system does not show if an action worked or failed. Example: clicking Submit and the screen freezes with no confirmation.
Inconsistency	Different parts of the app look or act differently. Example: a magnifying-glass icon for Search on one page, a binocular icon on another.
Accessibility gaps	Design ignores people with disabilities. Example: a video tutorial platform without captions for deaf users.
Compatibility issues	Interface fails on different devices or browsers. Example: a website that works on a laptop but crashes on a mobile browser.
Contextual neglect	The system does not adjust to the user's situation. Example: a map app with no dark mode, blinding the driver at night.
High cognitive load	Users must memorize too much or follow overly complex steps. Example: a registration system forcing you to remember three different SMS codes at once.
Trust deficiency	The system feels unsafe or hides security cues. Example: an online store asking for credit cards but lacking an SSL/HTTPS lock icon.

Remember: Eight problems in two families: (1) Screen problems — complexity, inconsistency, accessibility, compatibility, context. (2) Mind-and-trust problems — poor...

Exam: Common 8-mark question: "Describe problems of HCI with examples." Aim for at least six of the eight, each in the pattern Name → one-line meaning → example. Memorise the...

ایچ سی آئی مسائل وہ رکاوٹیں ہیں جو سسٹم کو انسانی ضرورت یا ذہنی حد کے مطابق نہ ہونے پر پیدا ہوتی ہیں۔

1.8 Methods for Improving HCI

Once you can name an HCI problem, you fix it with a matching method.

- Enhance usability: simplify navigation (a clear "Buy Now" button).
- Strengthen accessibility: support disabilities (captions on videos).
- Ensure consistency: keep layouts uniform (the same "Save" button everywhere).
- Provide clear feedback: show action results ("Payment Successful" popup).

Problem (1.7)	Method (1.8)
Cannot find "Send Money"	Enhance usability — one clear button
No captions on help video	Strengthen accessibility
Search icon changes from page to page	Ensure consistency
Submit freezes with no message	Clear feedback — "Payment Successful"
Portal crashes on mobile	Improve adaptability — responsive layout
Built features nobody asked for	UCD — interview, then pilot-test
Elderly user cannot type a long number	Multimodal — voice + touch
Complaints ignored after launch	Continuous testing and updates

Remember: Think "problem → method" as a pair. Complexity ↔ usability. No captions ↔ accessibility. Different icons ↔ consistency. Freeze ↔ feedback. Phone crash ↔...

ایچ سی آئی بہتر بنانے کے طریقے: نیویگیشن آسان کرنا، رسائی بڑھانا (کیپشن)، ایک جیسا لے آؤٹ، واضح فیڈبیک ("ادائیگی کامیاب")، ریسپانس...

1.9 User Interface Design (UI vs UX) ★ Golden

UI design is the process of creating and organizing the visual elements of a software application so users can interact with it easily and efficiently.

- 1.9.1 Components of a user interface ComponentRole ButtonsClickable elements that perform actions (Submit, Cancel).
- MenusLists of options for navigating the system.
- IconsSmall symbols representing tools or actions.

Component	Role
Buttons	Clickable elements that perform actions (Submit, Cancel).
Menus	Lists of options for navigating the system.
Icons	Small symbols representing tools or actions.
Text fields	Boxes where users type information (passwords, search).
Checkboxes / radio buttons	Selecting options in a form. Checkboxes allow several choices; radio buttons allow only one in a group.
Sliders	Adjusting values along a bar (volume, brightness).

Remember: Clothes vs day out. UI is the outfit (colours, buttons, icons). UX is the whole day (Did you arrive on time? Did you get wet? Were you stressed?). Examiners...

Exam: Highest-yield short question in the chapter: "Differentiate between UI and UX with an example." Write: (1) the two definitions, (2) the sentence "UI is what you see; UX..."

یو آئی ڈیزائن بصری عناصر — بٹن، مینو، آئیکن، ٹیکسٹ فیلڈ، چیک باکس/ریڈیو، سلائیڈر — کو ترتیب دینا ہے۔

1.9.3-1.9.5 Wireframing, Figma and Prototypes

1.9.3 Wireframing is creating a basic visual outline or blueprint of a user interface before the final design.

Lo-Fi sketch → Add details → Hi-Fi prototype

- It plans the structure without focusing on colours or detailed graphics.
- Low-fidelity wireframes (Lo-Fi) A simple black-and-white sketch showing basic layout (boxes and lines).
- Used early to test and share ideas quickly.

Remember: Lo-Fi = boxes. Hi-Fi = almost the real app. Prototype = clickable. Figma can hold all three stages. Trigger = what the user does. Action = where the fake app...

وائر فریم یو آئی کا بنیادی خاکہ ہے، رنگوں سے پہلے۔

1.11 Test and Evaluate HCI (Evaluation Methods)

Testing checks if the product works properly (finding bugs).

- Evaluation determines whether it meets users' needs (usability).
- Together they improve the product before release.
- A school portal can be "bug-free" (testing passed) and still unusable if students cannot find the date sheet (evaluation failed).

Method	Description
SUS (System Usability Scale)	A standard 10-item questionnaire used to calculate a score for how usable a system is.
NPS (Net Promoter Score)	Asks users: "How likely are you to recommend this app to a friend?" Measures user loyalty.
Formative evaluation	Done during development to collect early feedback and improve the design continuously.
Summative evaluation	Done at the end of development to measure overall success against the original goals.
Expert reviews	Usability specialists review the app against established design standards (heuristics).

Remember: Formative = forming it (during). Summative = summing it up (end). SUS = 10 questions on usability. NPS = "Would you recommend it?" Expert = specialist vs...

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایویلیوایشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔

1.11.2 Testing Methods (Usability, A/B, Automated) ★ Golden

Testing involves observing how the system behaves under specific conditions.

- Usability testing: observing real users while they perform realistic tasks, to measure efficiency and satisfaction. If the user gets stuck,...
- Task-based testing: assigning a specific task (for example, "Find a product and add it to cart") and strictly measuring completion time,...
- Automated audits: software bots scan websites to detect technical issues such as broken links, slow loading speeds or low colour contrast.
- A/B testing: comparing two versions of an interface (A and B). Half the users see A, half see B. The one with more clicks or sign-ups wins.

Remember: Task-based three numbers: Time, Success, Errors. A/B: two versions, split users, the higher clicks/sign-ups win. Bot: links, speed, contrast. Load: many users...

Exam: Golden 8-mark question: "Explain testing methods used in HCI." Write seven short paragraphs or a seven-row table. For full marks, define usability testing (observe real...)

سات ٹیسٹنگ طریقے: یوزیبیلٹی (حقیقی صارف کو کام کرتے دیکھنا)، ٹاسک بیسڈ (وقت، کامیابی، غلطیاں ناپنا)، خودکار آڈٹ (بوت ٹوٹے لنک،

سسٹ...

Career Opportunities in UI/UX

The same HCI skills you practised on ATMs, JazzCash and school portals are real jobs.

- Each role uses a different slice of this chapter.

Career	What the person does
UX designer	Researches user needs to ensure the app is intuitive and satisfying.
UI designer	Designs the visual layout (colours, typography, buttons) using tools such as Figma.
Interaction designer	Focuses on the flow between screens, animations and gestures.
Usability researcher	Runs SUS/NPS tests and A/B tests with real users to find flaws.
Accessibility specialist	Ensures products meet inclusive standards for people with disabilities.

Remember: UI looks; UX asks; interaction connects screens; research measures; accessibility includes everyone.

اس باب سے ملازمتیں: یو ایکس ڈیزائنر صارف کی ضرورت تحقیق کرتا ہے، یو آئی ڈیزائنر فگما میں رنگ اور بٹن بناتا ہے، انٹریکشن ڈیزائنر اس...

Key terms

Term	Meaning	اردو
HCI	Human-Computer Interaction; designing technology that is easy, efficient and responsive to people. Also called CHI or MMI.	انسان اور کمپیوٹر کا باہمی تعامل
Haptics	Physical feedback you can feel, such as a vibration.	لمسی / ارتعاشی جواب
Traditional interaction	Using standard input/output such as mouse, keyboard, keypad or simple tap and swipe.	روایتی تعامل
Natural interaction	Interacting without mouse or keyboard, using voice, face or body gesture.	قدرتی تعامل
End user	A person who directly operates the system.	حتمی صارف
Stakeholder	A person or group with an interest in the system's outcome (manager, client).	فریق مفاد
GUI	Graphical User Interface using windows, icons and menus.	تصویری صارف انٹرفیس
CLI	Command-Line Interface using typed text commands.	کمانڈ لائن انٹرفیس
NUI	Natural User Interface using gestures, body movement or immersive environments (VR, Kinect).	قدرتی صارف انٹرفیس
System feedback	The computer's response that confirms or guides an action.	نظام کا جواب
User feedback	Input the user sends to the system (Submit click, star rating).	صارف کی رائے / ان پٹ
Accessibility	Making digital systems usable by all, including people with diverse abilities.	رسائی پذیری
Need analysis	Finding the gap between the current situation and the desired outcome before designing a UI.	ضروریات کا تجزیہ
Cognitive load	How much the user must remember or process at once.	ذہنی بوجھ
UCD	User-Centred Design: focus on real user goals and pilot-test.	صارف مرکز ڈیزائن
UI	The visual elements of an application (buttons, menus, icons, fields, sliders).	صارف انٹرفیس

Term	Meaning	اردو
UX	The overall smoothness, comfort and satisfaction of using a system.	صارف کا تجربہ
Wireframe	A basic outline of a UI that plans structure without colours or detailed graphics.	انٹرفیس کا خاکہ
Lo-Fi	Low-fidelity sketch of boxes and lines used early to test ideas.	کم تفصیل والا خاکہ
Hi-Fi	High-fidelity prototype that looks almost like the final product.	زیادہ تفصیل والا نمونہ
Prototype	An early interactive model used to test design ideas before writing code.	ابتدائی چلنے والا نمونہ
Figma	A free, web-based tool for UI design and prototypes.	فگما (ویب پر یو آئی ٹول)
SUS	System Usability Scale: a 10-item questionnaire that scores how usable a system is.	نظام کی استعمال پذیری کا پیمانہ
NPS	Net Promoter Score: how likely a user is to recommend the app to a friend.	تجویز کا درجہ (وفاداری)
Formative evaluation	Evaluation during development to improve the design continuously.	دورانِ تعمیر جانچ
Summative evaluation	Evaluation at the end, measured against original goals.	اختتامی جانچ
A/B testing	Showing version A to half the users and version B to the other half; the better-performing version wins.	اے/بی موازنہ ٹیسٹ
Automated audit	Bots scanning for broken links, slow pages or low contrast.	خودکار جانچ
Heuristic	An established design standard used in expert reviews.	مستند ڈیزائن اصول

Chapter in one look

- **HCI** (also CHI, MMI) designs technology to be easy, efficient and responsive to people, using five sensory channels: sight, touch (haptics), hearing, voice and spatial.
- **Traditional interaction** uses mouse, keyboard, keypad or simple tap/swipe; **natural interaction** uses voice, face or body gesture without a mouse or keyboard.
- HCI applications cover **health care, banking, education, and networking/communication**, each with a clear user benefit (remote care, quick secure money, better learning, global connection).
- Four core components: **user** (end user vs stakeholder), **computer system**, **user interaction**, and **task** (complexity, knowledge, time).
- Interaction types include direct manipulation, command-based, menu-driven, form fill, conversational, gesture, voice, context-aware, biometric, haptic and multimodal.
- Interfaces: GUI, CLI, touch, VUI, NUI. **Environment** is physical, social and technical. **System feedback** is the machine's reply; **user feedback** is what the person sends in.
- HCI matters for productivity, inclusivity, innovation, economic impact and ethical responsibility. Poor design (unclear medical buttons) creates risk.
- Five accessibility principles: high contrast, readable fonts, colour-blind friendly labels, inclusive navigation, multimodal presentation.
- **Need analysis** finds the gap before UI design: identify users, collect data, analyse, prioritise, develop solutions (banking case: FaceID, large buttons, local language).
- Eight HCI problems: complexity, poor feedback, inconsistency, accessibility gaps, compatibility, contextual neglect, high cognitive load, trust deficiency.
- Improve HCI with usability, accessibility, consistency, clear feedback, adaptability (responsive design), UCD and pilot tests, multimodal input, and continuous testing.
- **UI** is the visual components (buttons, menus, icons, fields, checkboxes, sliders); **UX** is how the whole experience feels. UI is what you see; UX is how it feels.
- Lo-Fi wireframes plan structure; Hi-Fi prototypes look real; **Figma** is a free browser tool; prototypes use Trigger (On Tap) and Action (navigate) before code.
- Evaluation (SUS, NPS, formative, summative, expert/heuristic reviews) checks needs; testing (usability, task-based, automated, A/B, compatibility, load, keyboard/screen-reader) checks behaviour under conditions.

CHAPTER 2

Computational Thinking & Algorithms

★ 6 Golden · 24 cards · Trace Tables · Comparing Trace Tables and Stepwise Reasoning · Number of Steps (Big O Notation) · Concept of Data Structure · Linear Data Structures: Stack · Linear Data Structures: Queue

2.1 Analyze algorithms for correctness

An algorithm is correct if its output matches the expected results for a given input.

Trace tables → **paper execution**, **line by line** → **Stepwise reasoning** → **logic blocks**, **overall state**

- Analysing correctness means showing that the algorithm produces the intended output for all valid inputs, not just one lucky example.
- Two methods are used in this course:

Remember: Remember: runs ≠ correct. A logical error still produces output — just the wrong output. That is why we analyse correctness on paper before we trust the code.

الگورتھم تب درست (Correct) سمجھا جاتا ہے جب دیے گئے ان پٹ پر اس کا آؤٹ پٹ وہی ہو جو ہونا چاہیے۔

2.1.1 Trace Tables ★ Golden

A trace table is a technique used to monitor how the values of variables change during the execution of an algorithm.

- Dry running: execute the algorithm line by line by hand.
- Debugging: find logical errors where the algorithm runs but the output is wrong.
- Validation: check that the algorithm does what it is intended to do.
- Line / step number — which line is being executed.

```
AllHeights = [1.91, 2.01, 1.87, 1.57, 1.48]
Total = 0
for height in AllHeights:
    if height ≥ 1.57 and height ≤ 1.91:
        Total = Total + 1
print(Total)
```

Remember: How to fill a trace table without getting lost. Write a new row every time a variable changes or a condition is tested. Copy unchanged values into the new row...

Exam: Very common long question: "Dry-run the given algorithm using a trace table and state the output." Draw columns for every variable plus a condition column. Show the loop...

ٹریس ٹیبل الگورتھم کا کاغذی اجرا (Dry Run) ہے۔

2.1.2 Stepwise Reasoning

Stepwise reasoning (also called logic verification) breaks an algorithm into logical sections and checks the state of the algorithm at each stage.

- Analyse each step of the algorithm.
- Check: does it start in a valid state?
- Does it handle all possible cases?
- Are the conditions correct?

Check	What you verify
Initial state	Sum = 0 before the loop. Valid: nothing has been added yet.
Loop boundary	The loop runs exactly from $i = 1$ to $i = N$. No missing number, no extra number.
Iteration	Each i is added once: after k iterations, $\text{Sum} = 1+2+\dots+k$.
Final state	When the loop ends, Sum holds $1+2+\dots+N$, which is the required total.
Edge case $N = 0$	The loop body never runs, Sum stays 0. Correct (the sum of no numbers is 0).
Check case $N = 5$	Sum becomes 15, which matches $5 \times 6 / 2 = 15$.

```
N = 5
Sum = 0
for i in range(1, N + 1):
    Sum = Sum + i
print(Sum)
```

Remember: Think of stepwise reasoning as a teacher checking a proof, not as a cashier counting coins one by one. You ask "is this block always true?" instead of "what is..."

مرحلہ وار استدلال میں الگورتھم کو منطقی حصوں میں توڑ کر ہر مرحلے پر اس کی حالت (State) دیکھی جاتی ہے: کیا شروع درست ہے، کیا تمام ص...

2.1.3 Comparing Trace Tables and Stepwise Reasoning ★ Golden

Both methods verify correctness, but they fit different situations depending on the complexity and length of the algorithm.

Feature	Trace table	Stepwise reasoning
Best for	Small algorithm blocks, complex loops, and beginners learning to track variables.	Large systems, architectural logic, and abstract mathematical proofs.
Effort required	High. Very tedious and impractical if a loop runs 1,000 times.	Moderate. Needs strong logical deduction and abstract thinking.
Primary goal	Catching syntax misinterpretations or specific arithmetic errors line by line.	Catching conceptual or structural flaws in the overall design.
Level	Micro-level (line by line). Checks exact values.	Macro-level (logic blocks). Checks overall state.

Remember: Micro vs macro. Trace table = microscope (every cell). Stepwise reasoning = map (whole city). Beginners start with the microscope; designers need the map.

Exam: Differentiate questions are almost guaranteed: "Compare a trace table with stepwise reasoning." Write a four-row table (best for, effort, primary goal, micro/macro). Add...

ٹریس ٹیبل خرد بینی (Micro) سطح پر کام کرتا ہے: ہر لائن اور ہر قیمت۔

2.2 Evaluating the Clarity of an Algorithm

Clarity means an algorithm is easy to read, understand and modify.

Clarity → Modularity (parts) → Readability (text and flow)

- In many cases, evaluating clarity is even more important than making the algorithm slightly faster.
- A clear algorithm is easier to maintain, debug and extend over time.
- The two key aspects of clarity are:

Remember: Ask two questions: "Can I split this into parts?" (modularity) and "Can another student read it aloud and understand it?" (readability).

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔

2.2.1 Modularity

Modularity means breaking an algorithm into smaller, independent parts such as functions, classes or modules.

- Single Responsibility Principle (SRP). Each module should focus on one specific task. If a function calculates a sum, prints a report and...
- Coupling. Modules should be reusable in different contexts. Good modularity means loose coupling: a change in one part does not smash the...
- Abstraction. A well-designed algorithm hides complex implementation details. You can call a sorting function without knowing whether it...

Remember: SRP in one line: one function, one job. If you cannot name the function with a short verb-phrase (CalculateSum, FindAverage), it is doing too much.

ماڈیولریتی کا مطلب الگورتھم کو آزاد حصوں (فنکشن، کلاس، ماڈیول) میں توڑنا ہے۔

2.2.2 Readability

Readability is how easily another person can understand the purpose and logic of an algorithm without running it.

Names → Self-doc → Simple flow → Consistent

- Naming conventions — consistent rules for names. Good names make an algorithm easier to search and understand. Bad: $v = d / t$ Good:...
- Self-documentation — the algorithm explains itself, so you need fewer comments. Meaningful names and clean formatting make the purpose...
- Control-flow simplicity — keep the logic straightforward. Avoid deep nesting (IF inside IF inside IF), limit extra conditions, and keep...
- Consistency — one style throughout: indentation, spacing, and casing (camelCase or snake_case). Then the logic stands out instead of the...

```
# Hard to read
d = 100
t = 4
s = 20
v = d / t
if v > s:
    if t > 0:
        print(v)
```

Remember: Remember readability as N-S-S-C: Names, Self-documentation, Simple flow, Consistency.

پڑھائی پن یہ ہے کہ دوسرا شخص الگورتھم چلائے بغیر اس کا مقصد سمجھ لے۔

2.3 Assess Algorithm Efficiency

Algorithm efficiency is measured in terms of time and space — how fast it runs and how much memory it uses.

Time (speed) → Space (memory) → Efficiency

- Number of steps (described with Big O notation).
- Number of conditions (IF decisions).
- Number of repetitions (loops).

Remember: Efficiency is always "as the input grows". An algorithm that feels instant on 10 items can freeze on 10 million items if it is $O(n^2)$.

کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔

2.3.1 Number of Steps (Big O Notation) ★ Golden

When we discuss efficiency we ask how performance changes as the input size n increases.

- Big O notation is a mathematical method used to describe how the running time (or memory) grows as n grows.
- We ignore small constants and keep the most important term: if an algorithm does $3n + 5$ steps we write $O(n)$.
- In computer science, $\log n$ almost always means log base 2 (how many times you can halve n until you reach 1).

Class	Name	What happens when n grows	Typical example	Grade
$O(1)$	Constant	Time stays the same for 10 items or 10 million.	Access <code>arr[0]</code> (or any known index).	Excellent
$O(\log n)$	Logarithmic	Time grows very slowly; the problem is halved each step.	Binary search on a sorted list.	Great
$O(n)$	Linear	If n doubles, time roughly doubles.	Linear search of an unsorted list.	Fair / Good
$O(n \log n)$	Linearithmic	A little worse than linear; much better than n^2 . Common in good sorting.	Merge Sort (split in half, then merge).	Good
$O(n^2)$	Quadratic	Time grows with the square of n . Nested loops. Painful for large n .	Bubble Sort; compare every item with every other item.	Poor

```
arr = [23, 76, 12, 90, 34, 68]
print(arr[0])
```

Remember: Order from fastest to slowest (for large n): $O(1) \rightarrow O(\log n) \rightarrow O(n) \rightarrow O(n \log n) \rightarrow O(n^2)$. Mnemonic: "Constant Logs Never Need Squares" — Constant, Logarithmic,...

Exam: Memorise the five rows of the class table (name, growth, one example, grade). Typical MCQ: "Accessing `arr[0]` is ...?" $\rightarrow O(1)$. "Binary search is ...?" $\rightarrow O(\log n)$. "Nested..."

بگ او (Big O) بتاتا ہے کہ ان پٹ n بڑھنے پر وقت یا میموری کیسے بڑھتی ہے۔

2.3.2 Number of Conditions

Conditions (IF statements) are decision points.

- Computational cost. Each IF is extra work, even if the rest of the algorithm is simple.
- Nested conditions. IF inside IF often means a brute-force approach. It hurts both readability and speed.
- Short-circuiting (the optimisation goal). Stop evaluating as soon as the result is known. In a logical AND, if the first part is False the...

```
x = 0
y = 10
if x != 0 and (y / x) > 2:
    print("ratio is large")
else:
    print("skipped safely")
```

Remember: Put the cheapest / most likely False test first in an AND, and the most likely True test first in an OR. That is short-circuiting used on purpose.

ہر IF پروسیسر سے True/False پوچھتی ہے اس لیے زیادہ شاخیں کارکردگی گھٹاتی ہیں۔

2.3.3 Number of Repetitions

Repetitions (loops) are the most significant factor in algorithm efficiency.

- The structure of the loop decides how time grows with n .

Type of repetition	Conceptual impact	Efficiency grade
Linear loop	Processes each element exactly once. Time grows in proportion to size.	Good — $O(n)$
Logarithmic repetition	Reduces the problem size by half each iteration (binary search). Huge time savings.	Excellent — $O(\log n)$
Nested loops	For each element, process all other elements. Operations explode.	Poor — $O(n^2)$

Remember: Find the deepest loop. One loop $\approx O(n)$. Loop inside a loop $\approx O(n^2)$. A loop that halves $n \approx O(\log n)$.

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔

2.3.4 The Efficiency Evaluation Framework

To evaluate an algorithm at a conceptual level, ask three questions: Does it scale?

Does it scale? → Where is the bottleneck? → Is the extra complexity worth it?

- Does it scale? If n grows 10 times, does time jump 10 times (linear) or 100 times (quadratic)? Quadratic scaling fails for large datasets.
- What is the bottleneck? Focus on the most time-consuming section (usually the deepest loop) and try to improve that part.
- Is the cost worth it? A theoretically faster $O(\log n)$ algorithm can be too complex to code and maintain. For a list of just 50 items, a...

Situation	Scale	Bottleneck	Worth the cost?	Choice
50 unsorted class names	Tiny n	A single linear scan	Binary search needs a sort first, which is extra work and extra code	Linear search $O(n)$
1,000,000 sorted account numbers, searched all day	$n \times 10$ compared with 100,000 is already huge; quadratic would be impossible	The search loop	Yes — binary search is standard and well understood	Binary search $O(\log n)$

Remember: Framework mnemonic: S-B-W — Scale, Bottleneck, Worth it.

کارکردگی کے تین سوال: کیا الگورتھم بڑے n پر چل سکتا ہے (Scale)?

2.4 Refinements to improve clarity and efficiency

Improving an algorithm means reviewing it step by step: simplifying logic, removing redundant work, and picking smart data structures.

- Use meaningful names. Replace vague variables with descriptive ones. Bad: $a = b / c$. Good: $\text{average} = \text{totalSum} / \text{totalItems}$.
- Break into modules. Divide a massive block into functions, for example separating `CalculateSum()` from the printing logic.
- Improve structure and indentation. Packed loops and IF statements confuse the reader. Spacing reduces mental effort.
- Add comments. Keep them short. Explain why something is done, not just what is happening, especially for tricky logic.

Remember: Clarity: N-M-I-C (Names, Modules, Indentation, Comments). Efficiency: S-A-C-D (Shrink steps, Algorithm choice, Cache, Data structure).

وضاحت بہتر کرنے کے لیے معنی خیز نام، ماڈیول، درست indentation اور مختصر تبصرے استعمال کریں۔

2.5 Concept of Data Structure ★ Golden

A data structure is a systematic way of organising, storing and managing data so that an algorithm can access, modify and process it efficiently.

- Books on a shelf are arranged in a set order, so you can find a title quickly.
- Files in folders are grouped on purpose.
- Data structures do the same job inside computer memory.

Remember: Ask: "Does each item have at most one next item?" Yes → linear. "Can one item link to many, or can links form a net?" → non-linear.

Exam: Define data structure in full (organise, store, manage; efficient access and processing). Then split linear vs non-linear with the "one predecessor and one successor"...

ڈیٹا سٹرکچر ڈیٹا کو منظم طریقے سے رکھنے کا نظام ہے تاکہ الگورتھم اسے جلدی استعمال کر سکے۔

2.5.1 Linear Data Structures: Array

An array holds a collection of similar items (the same data type) placed next to each other in contiguous memory (one continuous block).

- Each item has a position number called an index, usually starting from 0.
- Because the address of any cell can be calculated from the index, you can jump straight to it.
- That is direct access (also called random access), and it is $O(1)$.

```
A = [23, 76, 12, 90, 34, 68]
print(A[3])
print(A[0])
```

Remember: Index 0 is the first shelf on the row. The third item that humans call "number 4" is index 3. Formula in your head: position from the left, minus one.

ایرے ایک جیسے عناصر کو میموری کے لگاتار بلاک میں رکھتا ہے۔

Linear Data Structures: Linked List

A linked list stores data as a chain. Unlike an array, the items are not one continuous block.

- the actual data;
- a link (pointer) holding the memory address of the next node.

```
nodes = [23, 76, 12, 90]
for data in nodes:
    print(data, end=" → ")
print("NULL")
```

Remember: Array = row of lockers with numbers on the doors. Linked list = treasure hunt: each clue (link) tells you where the next box is.

لنکڈ لسٹ زنجیر کی طرح نوڈز پر مشتمل ہے جو میموری میں بکھرے ہو سکتے ہیں۔

Linear Data Structures: Stack ★ Golden

A stack is a linear data structure that works on the LIFO principle: Last-In, First-Out.

- PUSH — add an item to the top.
- POP — remove the item from the top.

Step	Operation	Stack (bottom → top)	TOP	Popped value
0	(empty)	[]	none	
1	PUSH Data1	[Data1]	Data1	
2	PUSH Data2	[Data1, Data2]	Data2	
3	PUSH Data3	[Data1, Data2, Data3]	Data3	
4	POP	[Data1, Data2]	Data2	Data3
5	POP	[Data1]	Data1	Data2

```
stack = []
stack.append("Data1")
stack.append("Data2")
stack.append("Data3")
print("after 3 PUSHes:", stack)
print("POP:", stack.pop())
print("POP:", stack.pop())
print("left:", stack)
```

Remember: LIFO = Last In First Out = "last plate washed is first plate used" on a canteen pile. Undo, Redo, Back button and function calls are all stacks.

Exam: Define stack + LIFO, define PUSH and POP, mention TOP, give the Undo example, and draw the vertical boxes with TOP labelled. Trace questions: "Show the stack after PUSH..."

سٹیک LIFO پر چلتا ہے: جو سب سے بعد میں آیا وہ سب سے پہلے نکلے گا۔

Linear Data Structures: Queue ★ Golden

A queue organises data on the FIFO principle: First-In, First-Out.

- ENQUEUE — add an element at the Rear (back).
- DEQUEUE — remove an element from the Front.

Step	Operation	Queue (Front → Rear)	Front	Rear	Removed
0	(empty)	[]	—	—	
1	ENQUEUE 55	[55]	55	55	
2	ENQUEUE 76	[55, 76]	55	76	
3	ENQUEUE 90	[55, 76, 90]	55	90	
4	ENQUEUE D	[55, 76, 90, D]	55	D	
5	DEQUEUE	[76, 90, D]	76	D	55

Remember: FIFO = First In First Out = a canteen line. Front = the person being served. Rear = the person who just joined.

Exam: Define queue + FIFO, define ENQUEUE (rear) and DEQUEUE (front), give the printer example, draw the labelled Front/Rear diagram, and trace a short sequence. Examiners mix...

کیو FIFO پر چلتی ہے: جو پہلے آیا وہ پہلے نکلے۔

2.5.2 Non-Linear Data Structures: Tree

A tree organises data in a hierarchy. Everything branches out from a single starting point called the root.

- Everything branches out from a single starting point called the root.
- It creates a strict one-to-many (parent-child) relationship.
- That is an excellent way to show how roles flow in an organisation.

Remember: Tree = family / folders / organogram. One root. Branches down. Parent-child, not a web of equals.

ٹری ڈیٹا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک جڑ (Root) سے پھوٹتی ہے۔

Non-Linear Data Structures: Graph

A graph organises data when things connect in a flexible, complex way.

- Nodes (vertices) — the items (cities, people).
- Edges — the connections (roads, friendships).

Remember: Tree $\subset$ Graph in shape-talk: a tree is a special graph with one root and no cycles. If you need cycles or many-to-many links, say graph.

گراف لچکدار روابط کے لیے ہے: کوئی سخت درجہ بندی نہیں، کوئی بھی نوڈ کسی سے جڑ سکتا ہے۔

2.5.3 Operations on Data Structures

Whatever the shape, you keep meeting the same five operations:

Operation	Meaning	Quick picture
Insertion	Adding a new item into the structure.	PUSH on a stack; ENQUEUE on a queue; putting 40 into an array.
Deletion	Removing an item that is no longer needed.	POP; DEQUEUE; deleting a node from a list.
Traversal	Going through each item one by one.	Walking HEAD $\rightarrow$... $\rightarrow$ NULL; visiting every tree node.
Searching	Finding a specific item.	Linear search in an array; following links until 87 is found.
Sorting	Arranging items in a proper order (ascending or descending).	Bubble Sort on an array of marks.

Remember: Memorise I-D-T-S-S: Insert, Delete, Traverse, Search, Sort.

ہر ڈیٹا سٹرکچر پر پانچ عمل ہوتے ہیں: داخل کرنا (Insertion)، حذف کرنا (Deletion)، ایک ایک کر کے گزرنا (Traversal)، تلاش (Searching)...

2.6 Identify data structures in problem scenarios

To pick a structure, first see how the data is organised.

- Storing student grades for a class $\rightarrow$ Array. You usually know how many students there are, and all grades are the same type. Indices give...
- Music playlist / browser history $\rightarrow$ Linked list. Every song knows what comes next. When one finishes, follow the pointer. Frequent...
- Undo or Redo $\rightarrow$ Stack (LIFO). Each typing or formatting action is PUSHed. Undo POPs the most recent action.
- Printer task scheduling $\rightarrow$ Queue (FIFO). Documents print in the exact order they were received. First-come, first-served.

If the data...	Then it is...	Recommended structure
Follows "next" or "previous" logic	Linear	Array, Linked list
Follows "parent" and "child" logic	Non-linear	Tree
Follows "first-come, first-served" logic	Linear	Queue
Follows "connections / relationships" logic	Non-linear	Graph
Needs to be accessed in the reverse order of entry	Linear	Stack

Remember: Decision order: reverse-of-entry? $\rightarrow$ stack. First-come? $\rightarrow$ queue. Parent-child? $\rightarrow$ tree. Many-to-many routes? $\rightarrow$ graph. Known list with indexes? $\rightarrow$ array. Lots of...

اگر ہر عنصر کا واضح اگلا/پچھلا ہے تو ڈھانچہ لکیری ہے؛ اگر شاخیں یا کئی روابط ہیں تو غیر لکیری۔

2.7 Trace data retrieval in Array, List and Queue

Tracing data retrieval means following the step-by-step path of how an algorithm finds or accesses data.

- Array, linked list and queue are all linear, but the effort to find an item is very different because of how they are organised inside.

Loop iteration	Index	Array access	Value
1	0	Array[0]	10
2	1	Array[1]	50
3	2	Array[2]	20

```
Array = [10, 50, 20, 70, 90, 30]
print("Direct Array[2] =", Array[2])
for i in range(3):
    print(i + 1, i, f"Array[{i}]", Array[i])
```

Remember: Array jumps, list walks, queue waits. If the exam asks "why is array retrieval different?", write direct access by index versus sequential links versus Front...

ایرے میں اگر انڈیکس معلوم ہو تو سیدھی چھلانگ لگتی ہے، جیسے `Array[2] = 20`.

Career Opportunities in Algorithms

The skills in this chapter — proving correctness, writing clear modules, judging Big O, and choosing data structures — are the daily tools of several computing careers.

Career	What the person does
Software Engineer	Builds and maintains software that relies on data structures and efficient algorithms to solve logic problems.
Backend Developer	Builds server-side logic where choosing the right structure (array, queue) directly affects web performance.
Algorithm Engineer	Specialises in optimising algorithms for search engines, finance or logistics.
Systems Analyst	Evaluates and improves the efficiency and clarity of existing computer systems in an organisation.
Competitive Programmer	Solves complex algorithmic problems under time limits, a common path into top tech roles.

Remember: Every career in the table uses the same three tests you learned: correct, clear, efficient — plus the right data structure.

اس باب کی مہارتیں سافٹ ویئر انجینئر، بیک اینڈ ڈیولپر، الگورتھم انجینئر، سسٹمز اینالسٹ اور کمپیٹیٹو پروگرامر کے کام میں روز استعمال...

Key terms

Term	Meaning	اردو
Correctness	The algorithm's output matches the expected results for valid inputs.	درستگی
Trace table	A paper dry-run table of steps, variables, I/O and conditions.	ٹریس ٹیبل
Dry run	Executing an algorithm by hand, line by line.	دستی اجرا
Stepwise reasoning	Checking each logic block's state; also called logic verification.	مرحلہ وار استدلال
Clarity	How easy an algorithm is to read, understand and modify.	وضاحت
Modularity	Breaking work into independent functions, classes or modules.	ماڈیولریتی
Single Responsibility Principle	Each module should do one specific task.	واحد ذمہ داری کا اصول
Coupling	How tightly modules depend on each other; loose coupling is the goal.	ماڈیولز کی جڑت
Abstraction	Hiding implementation details so a caller need not know the inner method.	تجريد
Readability	How easily another person understands the logic without running it.	پڑھائی پن
Efficiency	How well an algorithm uses time (speed) and space (memory).	کارکردگی
Big O notation	A description of how running time or memory grows as n grows.	بگ او نوٹیشن
$O(1)$	Constant time; e.g. accessing <code>arr[0]</code> .	مستقل وقت
$O(\log n)$	Logarithmic time; the problem is halved each step (binary search).	لا جارتھمک وقت
$O(n)$	Linear time; work grows in proportion to n (linear search).	لکیری وقت
$O(n \log n)$	Work about n times log n (Merge Sort).	این لاگ این وقت
$O(n^2)$	Quadratic time; nested loops (Bubble Sort).	مربع وقت
Bottleneck	The slowest, most time-consuming section, often the deepest loop.	رکاوٹ، سب سے سست حصہ

Term	Meaning	اردو
Short-circuiting	Stopping a boolean evaluation as soon as the result is known.	جلدی ختم ہونے والی شرط
Memoization / caching	Computing a fixed value once and reusing it instead of recalculating.	ایک بار حساب، بار بار استعمال
Data structure	A systematic way to organise, store and manage data for efficient use.	ڈیٹا سٹرکچر
Linear structure	A sequence; each middle item has one predecessor and one successor.	لکیری ڈھانچہ
Array	Same-type items in contiguous memory, accessed by index.	ایرے
Linked list	Nodes of data + pointer, starting at HEAD and ending at NULL.	لنکڈ لسٹ
Stack	LIFO structure: PUSH and POP at the TOP.	سٹیک
Queue	FIFO structure: ENQUEUE at Rear, DEQUEUE at Front.	کیو
Tree	Hierarchical non-linear structure with one root and parent-child links.	ٹری
Graph	Nodes (vertices) joined by edges, with no strict hierarchy.	گراف
Traversal	Visiting each item in a structure one by one.	گزرنا، طواف
Direct access	Jumping to an array index without walking earlier items.	براہ راست رسائی

Chapter in one look

- **Correctness:** output matches expected results for valid inputs. Test with a trace table (manual dry run) or stepwise reasoning (logic verification).
- A **trace table** is micro-level: line, variables, I/O, conditions. Best for small loops. The soldier heights [1.91, 2.01, 1.87, 1.57, 1.48] print Total = 3.
- **Stepwise reasoning** is macro-level: valid start, all cases, correct conditions, loop stays correct. Sum 1..N starts at 0; N = 0 stays 0; N = 5 becomes 15.
- **Clarity** (read, understand, modify) often beats a tiny speed gain. It has two parts: modularity and readability.
- **Modularity:** SRP (one job), loose coupling, abstraction (call Sort without knowing Bubble vs Selection). Split CalculateSum and FindAverage.
- **Readability:** names (`velocity = distance / time`), self-documentation, simple flow, consistent style.
- **Efficiency** = time + space. Drivers: number of steps (Big O), conditions, repetitions.
- **Big O:** O(1) constant (`arr[0]`), O(log n) binary search, O(n) linear search, O(n log n) Merge Sort, O(n²) nested loops / Bubble Sort. Quadratic fails as n grows.
- **Framework:** Does it scale? What is the bottleneck (deepest loop)? Is the extra complexity worth it? For n = 50, simple O(n) can beat fragile O(log n).
- **Refinements:** meaningful names, modules, indentation, why-comments; shrink Bubble Sort's inner loop; better algorithms; cache constants; pick the right structure.
- **Data structure:** systematic organising of data. Linear (one after another): array, linked list, stack, queue. Non-linear: tree, graph.
- **Array:** contiguous, index from 0, direct access O(1), shifting on mid-insert. **Linked list:** nodes of data+link, HEAD to NULL, sequential access.
- **Stack LIFO:** PUSH/POP at TOP (Undo). **Queue FIFO:** ENQUEUE rear, DEQUEUE front (printer).
- **Tree:** one root, parent-child (college chart). **Graph:** vertices + edges, no strict hierarchy (GPS, friends).
- Five operations: insert, delete, traverse, search, sort. Choose by next/previous, parent/child, FIFO, relationships, or reverse-of-entry.
- **Retrieval:** array can jump to an index; a list walks from HEAD; a queue only removes the Front.

CHAPTER 3

Programming Fundamentals

★ 6 Golden · 18 cards · Lists (Ordered, Mutable, Methods) · Tuples (Ordered, Immutable) · Dictionaries (Key-Value Pairs) · Functions in Python · & 3.4 File Handling and Operations (Modes r, w, a) · The with Statement (Context Manager)

3.1 Data Structures in Python

A data structure is a way of organising values in a program so you can store them, find them and change them.

- Primitive data structures — the basic types that hold one value. Examples: integers (int), floating-point numbers (float), and booleans...
- Abstract data structures — higher-level containers built from primitive types so you can hold many values and run richer operations....

Category	Description	Examples
Primitive	Basic types that represent a single value.	int, float, bool
Abstract	Higher-level structures for collections and complex operations.	Lists, tuples, sets, dictionaries

```
age = 17
height = 5.8
passed = True
marks = [80, 75, 90]
print(type(age))
print(type(height))
print(type(passed))
print(type(marks))
```

Remember: Primitive = one piece. Abstract = a box of pieces. Think of a single rupee coin (primitive) versus a wallet that can hold many coins (list, tuple, set or...

ڈیٹا سٹرکچر (Data Structure) پروگرام میں ڈیٹا کو ترتیب دینے کا طریقہ ہے۔

3.1.1 Lists (Ordered, Mutable, Methods) ★ Golden

A list is Python's most versatile built-in data structure.

- It stores an ordered collection of items in a single variable.
- Lists are mutable (you can change them after you create them) and they are written with square brackets [].
- One variable can hold thousands of values.

After this line	List in memory
create	['apple', 'banana', 'apple']
.append('grape')	['apple', 'banana', 'apple', 'grape']
.remove('apple')	['banana', 'apple', 'grape'] — only the first apple goes
.sort()	['apple', 'banana', 'grape']

```
numbers = [10, 2, 15, 14, 3, 33]      # list of integers
names = ["Hassan", "Qasim", "Malika"] # list of strings
mixed = [1, "Python", 3.14, True]     # mixed data types
print(numbers)
print(names)
print(mixed)
```

Remember: Always remember [-1]. If a list has 10,000 items and you do not want to compute len(list) - 1, write list[-1]. It grabs the last item in one step.

Exam: Lists appear every year: "What is a list?", "List four methods with one line each", "Differentiate list and tuple", "Write a program to search an element", "Predict the..."

فہرست (List) مربع بریکٹ [] میں لکھی جاتی ہے، ترتیب برقرار رکھتی ہے اور تبدیل کی جا سکتی ہے (Mutable)۔

3.1.2 Tuples (Ordered, Immutable) ★ Golden

A tuple is an ordered collection that is immutable — once created, you cannot add, remove or replace items.

- Think of it as a read-only list.
- Tuples use parentheses ().
- Like lists, they allow duplicates, mixed types, and indexing from 0 (including negative indices).

Method	Meaning
<code>.count(value)</code>	How many times the value appears.
<code>.index(value)</code>	Index of the first occurrence.

```
t = (10, 20, 30, 40, 50)
print(t[0])
print(t[-1])
print(t[1:4])
print(t[::-1])
```

Remember: Tuple = locked lunchbox. You can see what is inside; you cannot swap the sandwich. Memory: "Tuple is Tight (immutable), written with round brackets that look...

Exam: Expect "Differentiate list and tuple" (at least four points: brackets, mutability, methods, speed/protection), "Why use a tuple?", and "What happens if you write `t[0] = ...`"

ٹیپل (Tuple) گول بریکٹ () میں لکھا جاتا ہے، ترتیب رکھتا ہے مگر تبدیل نہیں ہو سکتا (Immutable).

3.1.3 Sets (Unordered, Unique, Math Operations)

A set stores an unordered, unindexed collection of unique elements.

- It is inspired by mathematical sets.
- You write a set with curly braces {}.
- Duplicates are removed as soon as the set is created.

Method	Description
<code>.add(value)</code>	Adds one element.
<code>.update(values)</code>	Adds many elements from a list, tuple or set.
<code>.remove(value)</code>	Removes the value. Raises <code>KeyError</code> if it is missing.
<code>.discard(value)</code>	Removes the value. Does not crash if it is missing.
<code>.clear()</code>	Removes every element. Result is <code>set()</code> .

```
nums = {1, 4, 3, 3, 2, 2}
print(nums)
```

Remember: SET = Same-item Eaten Twice-never. Empty set is `set()`, never `{}`. Safe delete = `discard`; angry delete = `remove`.

سیٹ (Set) منفرد عناصر کا بغیر ترتیب اور بغیر انڈیکس مجموعہ ہے، جو {} میں لکھا جاتا ہے۔

3.1.4 Dictionaries (Key-Value Pairs) ★ Golden

A dictionary stores data as key-value pairs.

- Ordered (Python 3.7+): insertion order is kept.
- Mutable: you can add, change and delete pairs.
- Keys are unique: if you repeat a key, the last value silently overwrites the earlier one. Values may repeat.

Method	Returns
<code>.keys()</code>	All keys.
<code>.values()</code>	All values.
<code>.items()</code>	Key-value pairs as tuples.

```
d = {"name": "Alice", "age": 30}
print(d)
d["age"] = 31
print(d)
d["city"] = "NYC"
print(d)
del d["city"]
print(d)
```

Remember: Keys are unique, like roll numbers. Values are the student records and may repeat (two students can both score 80). Memory: "get is gentle, brackets can break."

Exam: High-yield: definition of a dictionary, difference between `d[key]` and `d.get(key)`, "what if a key is repeated?", traversal with `.items()`, and a full comparison table of...

ڈکشنری (Dictionary) کلید اور قدر (Key-Value) کے جوڑوں میں ڈیٹا رکھتی ہے۔

3.1.5 Common Built-in Functions (min, max, len, sum)

Several built-in functions work across lists, tuples, sets and dictionaries.

- You do not call them with a dot on the object (except when a type also has its own method, such as `list.sort()`).

Function	Purpose	Works on
<code>min()</code>	Smallest element (numeric or alphabetical)	List, tuple, set, dictionary keys
<code>max()</code>	Largest element	List, tuple, set, dictionary keys
<code>len()</code>	Number of elements (for a dict: number of keys)	List, tuple, set, dictionary
<code>sum()</code>	Adds numeric elements	List, tuple, set, dictionary values via <code>sum(d.values())</code>
<code>sorted()</code>	Returns a new sorted list ; original unchanged	List, tuple, set, dictionary keys

```
s = {5, 1, 9, 3, 7}
print(len(s))
print(min(s))
print(max(s))
print(sum(s))
print(sorted(s))
```

Remember: LEN-MIN-MAX-SUM — count, smallest, largest, total. For a dictionary: `len/min/max/sorted` use keys; `sum` needs `.values()`.

`len` () عناصر گنتا ہے، `min` () سب سے چھوٹا، `max` () سب سے بڑا، `sum` () اعداد کا مجموعہ، اور `sorted` () نئی ترتیب شدہ فہرست دیتا ہے بغیر اصل کو بدلے ہیں۔

3.1.6 Activity: Student Attendance System

Teachers need a quick way to mark who is in class.

Store names in a list → Ask present? for each → Count yes → Print totals

- This program stores names in a list, asks about each student, counts how many said yes, and prints a summary with `len()`.

Student	Typed answer	After <code>lower()</code>	<code>present_count</code>
Ali	yes	yes → add 1	1
Sara	no	not yes	1
Ayesha	yes	yes → add 1	2

Remember: Counter pattern: start at 0, loop, add 1 when a condition is true. You will reuse this in the word-frequency program (add 1 per word) and the grade tracker...

حاضری کا پروگرام طلبہ کے نام ایک فہرست میں رکھتا ہے، ہر نام کے لیے پوچھتا ہے، اور `yes` پر شمار کار (Counter) ایک بڑھاتا ہے۔

3.1.7 Activity: Word Frequency Counter

Count how many times each word appears in a sentence.

- This is a classic use of a dictionary plus `.get()`: the word is the key, the count is the value.

After seeing	<code>word_freq</code>
python	<code>{'python': 1}</code>
is	<code>{'python': 1, 'is': 1}</code>
easy	<code>{'python': 1, 'is': 1, 'easy': 1}</code>
python (again)	<code>{'python': 2, 'is': 1, 'easy': 1}</code>
... end of sentence	<code>{'python': 3, 'is': 3, 'easy': 1, 'fun': 1, 'powerful': 1}</code>

Remember: The heart of every counter is one line: `d[item] = d.get(item, 0) + 1`. You can count letters, roll numbers or votes with the same pattern.

جملے کو `split` () سے الفاظ کی فہرست بناتے ہیں، پھر ہر لفظ ڈکشنری میں کلید ہوتا ہے اور اس کی گنتی قدر۔

3.2 Functions in Python ★ Golden

When the same instructions are written again and again, that is duplicate code.

def → **name** → **parameters** → **indented body** → **return (optional)**

- Duplicate code makes a program longer, harder to read, and painful to update (you must fix every copy).
- A function lets you write the instructions once and run them whenever you call the function's name.

Benefit	Meaning
Code reusability	Write once, use anywhere.
Modularity	Break a big problem into small named tasks.
Readability	Shorter, cleaner files with meaningful names.
Maintainability	Fix a bug in one place and every call is fixed.

```
def greet():  
    print("Hello, World!")  
  
greet()  
greet()
```

Remember: DRY — Don't Repeat Yourself. If you are about to copy-paste a block, wrap it in **def**. Name functions with verbs: **greet**, **add**, **save_grades**.

Exam: Write the four benefits (reusability, modularity, readability, maintainability) as separate bullets — that is a standard short question. For long questions, define a...

فونکشن (Function) ہدایات کا وہ بلاک ہے جو ایک بار لکھا جاتا ہے اور بار بار بلایا جا سکتا ہے۔

3.2.1 Types of Functions (Built-in vs User-defined)

There are two main types of functions in Python.

- a) Built-in functions are already defined by Python.
- Examples you have already used: `len()`, `range()`, `sum()`, `print()`, `input()`, `min()`, `max()`, `sorted()`, `int()`, `float()`, `type()`.
- b) User-defined functions are functions you create with **def** for tasks that no built-in covers in one step — greeting a user, dividing with...

	Built-in	User-defined
Who writes it?	The Python language	The programmer
How you start	Just call the name	First write <code>def name(...):</code>
Examples	<code>len("Ali")</code> → 3	<code>def add(a, b): return a + b</code>
Purpose	Everyday operations needed by everyone	Your program's specific jobs

```
print(len("Sindh"))  
  
def greet_student(name):  
    print("Hello,", name)  
  
greet_student("Hassan")  
greet_student("Malika")
```

Remember: If Python already has it (`len`, `sum`, `max`), do not rewrite it. Write a user-defined function when you are packaging your steps.

بلٹ ان فونکشن پائتھن میں پہلے سے موجود ہوتے ہیں، جیسے `len()` اور `print()`۔

3.2.2 Functions with Return Values

A function can either print a result (show it on the screen) or return a result (send it back to the caller).

- A returned value can be stored in a variable, printed later, or used in a further calculation.
- The call is replaced by the returned value, as if you had written that value in its place.

```
def multiply(a, b):  
    return a * b  
  
result = multiply(5, 3)  
print("The answer is:", result)
```

Remember: **return** = send back; **print** = show now. **return** also exits the function immediately. Lines after **return** in the same function do not run.

return نتیجہ کال کرنے والے پروگرام کو واپس بھیجتا ہے تاکہ وہ متغیر میں محفوظ ہو سکے یا آگے حساب میں لگے۔

3.2.3 Activity: Arithmetic Calculator Using Functions

Each arithmetic operation is a separate function.

- That is modularity: addition is one small task, division is another (with a guard against dividing by zero).
- The main program only asks for two numbers and calls the functions.

Remember: Four functions, four names: add, subtract, multiply, divide. The main program should look like a contents page — a sequence of calls — not a pile of formulae.

کیلکولیٹر میں ہر عمل الگ فنکشن ہے: جمع، تفریق، ضرب، تقسیم۔

3.2.4 Scope of Local and Global Variables

Scope is the region of a program where a variable can be seen and used.

- Python has two scopes you must know for this chapter.
- A local variable is created inside a function.
- It exists only while that function is running.

```
def calc():
    secret = 50
    print(secret)

calc()
print(secret)
```

Remember: Local = inside the room; global = in the corridor. Everyone can see the corridor. Things left in a room are gone when you leave. Prefer locals. Too many...

دائرہ کار (Scope) بتاتا ہے کہ متغیر کہاں دکھائی دیتا ہے۔

3.3 & 3.4 File Handling and Operations (Modes r, w, a) ★ Golden

File handling is a program's ability to create, read, write, update and delete files on disk.

- Variables live in RAM and vanish when the program ends.
- Files live on a hard disk or SSD and stay there — they are persistent.

Type	What it stores	Examples
Text file	Human-readable characters. You can open it in a text editor.	.txt, .csv, .py
Binary file	Raw bytes, not meant to be read as text.	.jpg, .mp3, .dat

```
new_file = open('myfile.txt', 'w')
new_file.write("The quick brown fox jumps over the lazy dog!!!\n")
new_file.close()
print("File has been created successfully.")
```

Remember: r = Read existing, w = Wipe then write, a = Add at end. Remember: write does not add a newline unless you type \n.

Exam: Define file handling, contrast RAM with disk, list modes r / w / a with one property each, and warn that w overwrites. Short programs that write one line and close the...

فائل ہینڈلنگ سے ڈیٹا ڈسک پر مستقل (Persistent) رہتا ہے، برخلاف RAM کے متغیرات کے جو پروگرام بند ہوتے ہی مٹ جاتے ہیں۔

3.6 The with Statement (Context Manager) ★ Golden

It is easy to forget .close(). If the program crashes before .close() runs, the file may be incomplete or corrupted.

with open(...) as f → use f → block ends → file closed automatically

- It is easy to forget .close().
- If the program crashes before .close() runs, the file may be incomplete or corrupted.
- Python's with statement is a context manager: it opens the file, runs the indented block, and always closes the file when the block ends —...

```
with open('myfile.txt', 'w') as new_file:
    new_file.write("Context managers are safe!\n")
# no .close() here — with already closed the file
```

Remember: with = "take, use, put back." Prefer with open(...) as f: in every new program. You still choose the mode: 'r', 'w' or 'a'.

Exam: Define the with statement as a context manager that guarantees close. Contrast it with manual open/close. Be ready to write a four-line program that writes with with and...

with بیان ایک کانٹیکسٹ مینیجر (Context Manager) ہے جو فائل کو بلاک کے اختتام پر خود بند کر دیتا ہے، چاہے اندر غلطی ہو جائے۔

3.8 Errors and Exceptions (Try-Except)

An exception is an error raised while the program is running — for example opening a file that is not there.

- Without handling, Python prints a traceback and stops.
- A try/except block catches the error and lets you show a clear message instead of crashing.

Error type	Meaning
FileNotFoundError	The path does not exist.
PermissionError	The operating system will not let you read or write that file.
IOError	A general input/output problem (disk full, disk disconnected).

```
try:
    with open('missing_data.txt', 'r') as f:
        content = f.read()
        print(content)
except FileNotFoundError:
    print("Error: The file does not exist. Please check the name.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
```

Remember: try = attempt; except = backup plan. Catch the specific error first (FileNotFoundError), then a general Exception if you need it. Order matters: a bare except...

ایکسپشن (Exception) چلتے پروگرام کی غلطی ہے، جیسے غیر موجود فائل کھولنا۔

3.9 Mini File Handling Project: Grade Tracker

This project uses 'w' to create a grades file, 'a' to add a student later, and 'r' to read every line and compute an average.

Write names and marks ('w') → Append a new student ('a') → Read lines ('r') → Split, total, average

- It combines file handling, string methods and arithmetic.

Remember: Same file, three modes, three with blocks. Never open the same file in 'w' when you only meant to add David — that would delete Alice, Bob and Carol.

گریڈ ٹریکر پہلے 'w' سے فائل بناتا ہے، پھر 'a' سے نیا طالب علم جوڑتا ہے، پھر 'r' سے تمام سطریں پڑھ کر اوسط نکالتا ہے۔

Career Opportunities

The skills in this chapter — data structures, functions and files — are the same skills used in many jobs.

- Python / software developer — builds applications, scripts and tools in Python, one of the most widely used languages in industry.
- Data analyst — uses lists and dictionaries, then file handling, to process, visualise and interpret data for decisions.
- Automation engineer — writes scripts that repeat file and testing tasks so humans do not have to.
- Game developer — uses functions, data structures and logic to build interactive games.

Remember: Four careers: P-D-A-G — Python developer, Data analyst, Automation engineer, Game developer.

اس باب کی مہارتیں پائتھن ڈویلپر، ڈیٹا اینالسٹ، آٹومیشن انجینئر اور گیم ڈویلپر کے کام میں روز استعمال ہوتی ہیں۔

Key terms

Term	Meaning	اردو
Data structure	A way of organising values so a program can store and use them.	ڈیٹا سٹرکچر
Primitive type	A basic single value: <code>int</code> , <code>float</code> , <code>bool</code> .	بنیادی قسم
List	Ordered, mutable sequence written with <code>[]</code> .	فہرست
Index	The position of an item, starting at 0 (or -1 from the end).	انڈیکس
Slicing	Copying a portion of a sequence with <code>start:stop:step</code> .	سلائسنگ
Traversal	Visiting each element in turn, usually with a loop.	گشت / تراورسل
Mutable	Can be changed after creation.	قابل تبدیلی
Immutable	Cannot be changed after creation.	ناقابل تبدیلی
Tuple	Ordered, immutable sequence written with <code>()</code> .	ٹیپل
Set	Unordered collection of unique items written with <code>{}</code> .	سیٹ
Union	All unique items from two sets (<code> </code>).	اجتماع
Intersection	Items common to both sets (<code>&</code>).	اشتراک
Dictionary	Mutable map of unique keys to values.	ڈکشنری
Key	The unique lookup name in a dictionary.	کلید
Value	The data stored under a key.	قدر
Function	A named, reusable block of code.	فنکشن
Parameter	An input name in a function definition.	پیرامیٹر
Return value	The result sent back with <code>return</code> .	واپسی کی قدر
Local variable	A name that exists only inside a function.	مقامی متغیر
Global variable	A name defined outside all functions.	عالمی متغیر
File handling	Creating, reading, writing and closing disk files.	فائل ہینڈلنگ
File mode	How a file is opened: <code>r</code> , <code>w</code> , <code>a</code> (or binary).	فائل موڈ
Context manager	A <code>with</code> block that sets up a resource and cleans it up.	کانٹیکسٹ مینیجر
Exception	A runtime error object, such as <code>FileNotFoundError</code> .	ایکسپشن
try / except	A pair of blocks that attempt code and handle errors.	ٹرائی / ایکسپٹ

Chapter in one look

- Python programs rest on primitive types (`int`, `float`, `bool`) and abstract structures: lists, tuples, sets and dictionaries.
- A **list** is ordered and mutable, written with `[]`. Index from 0; `[-1]` is last. Slice stop is exclusive. Methods include `append`, `insert`, `remove`, `sort`, `reverse`, `clear`.
- A **tuple** is ordered and immutable, written with `()`. Item assignment raises `TypeError`. Only `count` and `index` exist as methods.
- A **set** is unordered and unique, written with `{}`. Use `|` & `-` & `^` for union, intersection, difference and symmetric difference. Prefer `discard` over `remove`.
- A **dictionary** maps unique keys to values. `d[key]` can raise `KeyError`; `d.get(key, default)` is safe. Duplicate keys keep the last value.
- `min`, `max`, `len`, `sum` and `sorted` work across structures. On a dict, `min/max/sorted` use keys; total values with `sum(d.values())`. `sorted()` returns a new list; `.sort()` changes a list in place.
- Functions written with `def` give reusability, modularity, readability and maintainability. Built-in functions already exist; user-defined functions solve your tasks.
- `return` sends a value to the caller. Without `return`, the result is `None`. Local variables live inside a function; global variables live outside all functions.
- Files persist on disk. Modes: `r` read (must exist), `w` write (overwrites), `a` append. Methods: `read`, `readlines`, `write`, `close`.
- The `with` statement is a context manager that closes the file automatically. Loop line by line to read large files safely.
- `try/except FileNotFoundError` stops a missing file from crashing the program.
- The grade tracker writes with `w`, appends with `a`, reads with `r`, splits each `Name: mark` line, and computes the mean.

CHAPTER 4

Data and Analysis

★ 5 Golden · 9 cards · Data Sources and Database Connection · Pandas and DataFrames · Handling Missing Values (NaN) · Data Visualization and Types of Graphs · Descriptive Statistics

4.1 Concept of Data Analysis

Data analysis is examining, organising and interpreting data so you can extract useful information.

- Schools study results, shops study sales, hospitals study patient records — all to make a decision, spot a trend or solve a problem.
- Python is popular for this work because it is readable, talks to files and databases, does statistics, and draws graphs.
- Five lines of Pandas often replace an hour of spreadsheet clicking.

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔

4.1.1 – 4.1.2 Data Sources and Database Connection ★ Golden

Data sources are where information comes from: school records, websites, sensors, databases.

Python → Driver → Connection → Cursor → SQL → Database

- Connectivity is the conversation between your Python program and that source so you can read and update data.
- Five components of a database connection, in the order they actually work:

#	Component	Role
1	Database	Organised store (SQLite = one .db file; MySQL = a server for big apps)
2	Driver	Translator between Python and that DBMS (sqlite3 is built in)
3	Connection object	Opens, manages and closes the pipeline
4	Cursor object	Runs SQL and walks through result rows
5	SQL queries	CREATE, INSERT, SELECT, UPDATE, DELETE

Exam: Write the five names in order and one line each. A common trap is swapping cursor and connection: the connection is the pipe; the cursor is the messenger that travels...

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔

4.2 Creating a SQLite Database in Python

SQLite is built into Python. The whole database is one .db file next to your script — no password, no port, no server.

- SQLite is built into Python.
- The whole database is one .db file next to your script — no password, no port, no server.
- Nine steps you must be able to write from memory:

Step	Why it matters
connect	Creates the file if it does not exist
cursor	Vehicle that carries SQL
execute	Runs one SQL command
commit	Writes INSERT/UPDATE/DELETE to disk
fetchall	Brings SELECT results into a Python list of tuples
close	Frees memory and unlocks the file

Remember: Strings inside SQL use single quotes: 'Ali'. Python strings around the SQL use double quotes (or a triple-quoted string).

ایس کیو لائٹ پائتھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔

4.3 Pandas and DataFrames ★ Golden

Pandas is a third-party library for tables.

- Install once: pip install pandas in the terminal (not inside the .py file).
- Convention: import pandas as pd.
- A DataFrame is a 2-D table of rows and columns, like Excel or an SQL table.

```
import pandas as pd
data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Math": [85, 90, 78],
    "Science": [80, 88, 70],
}
df = pd.DataFrame(data)
print(df)
```

پینڈا ڈیٹا فریم دیتا ہے۔

4.3.3 Handling Missing Values (NaN) ★ Golden

Missing cells are NaN ("Not a Number"). Statistics crash or lie if you ignore them.

- Statistics crash or lie if you ignore them.
- Three tools: isnull() (True where empty), fillna(value) (replace), dropna() (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

Trap: Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean).

Exam: State: identify with isnull/sum, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean...

خالی خانے NaN ہیں۔

4.4 Data Organization and Representation

Organised data is easier to read. Three everyday forms: Table — exact values side by side (Ali 85 Math, 80 Science).

- Table — exact values side by side (Ali 85 Math, 80 Science).
- Frequency distribution — how many items fall in each range (bin).
- Graph — a picture of a trend or comparison (next topics).

Range of marks	Frequency (students)
50–59	5
60–69	7
70–79	9
80–89	4
90–100	2

ٹیبل درست عدد دکھاتی ہے، تعدد کی جدول بتاتی ہے کتنے طلبہ کس رینج میں ہیں، گراف رجحان دکھاتا ہے۔

4.5 Data Visualization and Types of Graphs ★ Golden

Pick the graph that matches the question. Then label axes and give a title — an unlabelled chart scores zero in an exam.

- Then label axes and give a title — an unlabelled chart scores zero in an exam.

Chart	Shows	Watch out
Bar / column	Compare discrete categories (subjects, departments)	Column = vertical; bar often = horizontal
Line	Change over time (temperature, visitors)	Needs an ordered x-axis (months, years)
Pie	Parts of one whole as percentages	Bad with many or almost-equal slices
Histogram	Distribution of one numeric variable in bins	Bars touch (continuous range), unlike a bar chart
Scatter	Relationship of two numeric variables	Each dot is one pair (x, y)
Box plot	Min, Q1, median, Q3, max, and outliers	Best tool for spotting extreme values

```
import matplotlib.pyplot as plt
subjects = ["English", "Math", "Science", "History"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue", edgecolor="black")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks out of 100")
```

Exam: "Which graph and why?" is a 3-mark gift. Write the name, what it shows, and one reason the others are worse.

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکیتھر دو متغیرات کا تعلق، باکس پلا...

4.6 Descriptive Statistics ★ Golden

Descriptive statistics squeeze a long list into a few honest numbers.

- Central tendency asks "what is typical?".
- Spread asks "how mixed is the list?".

Measure	How	Use when
Mean	Sum ÷ n. Uses every value.	Values sit close together. Example: ordinary test scores.
Median	Middle after sorting. Odd n: centre. Even n: mean of the two centre values.	Outliers would wreck the mean (CEO salary).
Mode	Most frequent value. May be none, one, or several.	Most popular choice (shoe size).

Trap: Do not report variance as if it were in "marks".

Exam: Always state whether you used n or n-1. Show the sorted list for the median. For SD, show: mean → deviations → squares → divide → square root. A box plot pictures min,...

اوسط سب کا مجموعہ تقسیم تعداد، میڈین ترتیب کے بعد بیچ، موڈ سب سے زیادہ بار آنے والا۔

Career Opportunities

The skills in this chapter map onto real jobs: data analyst (Pandas, Excel), database administrator (SQLite/MySQL), data scientist (statistics + modelling), visualisation specialist (Matplotlib, Power BI),...

یہ باب ڈیٹا اینالسٹ، ڈیٹا بیس ایڈمن، ڈیٹا سائنسدان، ویژولائزیشن اور بیک اینڈ ڈویلپر جیسے کیریئر کی بنیاد ہے۔

Key terms

Term	Meaning	اردو
Data analysis	Examining data to extract useful information.	ڈیٹا کا تجزیہ
Connectivity	Link between a program and a data source.	رابطہ کاری
Driver	Translator between Python and a DBMS.	ڈرائیور
Connection	Open pipeline to the database.	کنکشن
Cursor	Object that executes SQL and fetches rows.	کرسر
SQLite	Built-in file-based database.	ایس کیو لائٹ
commit	Save INSERT/UPDATE/DELETE to disk.	کمیٹ / محفوظ کرنا
Pandas	Library for table-like data.	پینڈا
DataFrame	2-D labelled table of rows and columns.	ڈیٹا فریم
NaN	Missing value ("Not a Number").	خالی / غیر موجود قدر
Frequency	How many items fall in a class/bin.	تعدد
Histogram	Distribution chart whose bars touch.	ہسٹوگرام
Scatter plot	Dots showing two numeric variables.	اسکیٹر پلاٹ
Box plot	Min, Q1, median, Q3, max (and outliers).	باکس پلاٹ
Mean	Arithmetic average.	اوسط
Median	Middle value of a sorted list.	میڈین / درمیانی قدر
Mode	Most frequent value.	موڈ / کثرت
Range	Max – min.	حد / رینج
Variance	Mean of squared deviations from the mean.	ویریانس / تغیر
Standard deviation	Square root of variance (original units).	معیاری انحراف
Outlier	An extreme unusual value.	آؤٹ لائر

Chapter in one look

- Data analysis turns raw facts into decisions. Python reads files and databases, calculates and plots.
- Connection chain: database → driver → connection → cursor → SQL.
- SQLite: connect, cursor, execute, commit, fetchall, close. No commit means no save.
- A DataFrame is a 2-D table. Load with a dictionary, `read_csv` or `read_excel`. New columns are vectorised arithmetic.
- NaN: find (`isnull`), fill (`fillna` , often with the mean), or drop (`dropna`).
- Tables give exact values; frequency bins show how many in each range.
- Bar/column = categories; line = time; pie = parts of a whole; histogram = numeric distribution (touching bars); scatter = two variables; box = five-number summary + outliers.
- Mean uses every value and is sensitive to outliers; median is the middle; mode is the most common.
- Range = max–min. Variance = mean squared deviation ($\div n$ or $\div n-1$). SD = $\sqrt{\text{variance}}$. pandas `.std()` uses $n-1$.

CHAPTER 5

Application and Impacts of Computing

★ 5 Golden · 17 cards · Neural Networks (NN) · Components of Neural Networks · Secure Collaboration (Authentication, MFA, Access Control) · Common Security Threats (Malware, Phishing, DoS) · Functions of Equity and Equal Access

5.1 Machine Learning, Neural Networks and Deep Learning

Machine learning (ML), neural networks (NN) and deep learning (DL) are important areas of artificial intelligence (AI).

- Face unlock on smartphones.
- Speech recognition (for example Siri, Alexa, or Urdu voice typing on a phone).
- Photo search categorisation (typing "car" in a gallery and seeing car photos).
- Medical diagnosis from X-rays and scans.

Remember: Russian-doll memory trick. Picture three boxes. The outer box is AI. Inside it is a smaller box ML. Inside that is the smallest box DL. You cannot take DL out...

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔

5.1.1 Machine Learning (ML)

A machine learning model is the system that learns from data and gives an output.

1. Data input → 2. Learning (find patterns) → 3. Prediction on new data

- Think of it as the trained brain of a computer program.
- In ordinary programming you write every rule yourself: if marks ≥ 40 then "Pass".
- In machine learning you do not list every rule.

Remember: D-L-P. Machine learning is Data in, Learn patterns, Predict. "Data Learns to Predict."

مشین لرننگ ماڈل وہ نظام ہے جو ڈیٹا سے سیکھ کر نتیجہ دیتا ہے — کمپیوٹر پروگرام کا تربیت یافتہ دماغ۔

5.1.2 Neural Networks (NN) ★ Golden

A neural network is a computer model that learns patterns from data.

Input layer → Hidden layer(s) → Output layer

- Input layer — receives the raw data (pixels of a face, samples of a sound, marks of a student).
- Hidden layer(s) — the "thinking" in the middle. Nodes here combine the inputs and learn features (eye distance, nose shape, an edge in a...)
- Output layer — gives the final answer (unlock / keep locked; "this is a cat"; "pass / fail").

Remember: I-H-O. Input receives, Hidden thinks, Output decides. "In Hidden Out." A neural network is a brain-inspired maths machine, not a copy of a living brain.

Exam: Very common questions: "What is a neural network?" (2–3 marks), "Name its three layers and explain each with the face-unlock example" (5 marks), and "How is a neural..."

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔

5.1.3 Deep Learning (DL)

Deep learning is a machine-learning method in which computers use multi-layer neural networks to learn complex patterns from data.

- Early layers learn simple features — edges, shapes, colours.
- Deeper layers combine those simple features to understand complex patterns — a whole face, or a specific car model (a Cultus versus a...

	Neural network (the model)	Deep learning (the method)
What it is	A system/model that has layers, nodes and connections to help it learn.	A specific method that trains neural networks containing many hidden layers .
Depth	Can be simple (shallow) with just one or two hidden layers.	Always "deep," requiring huge amounts of data and processing power to learn complex patterns.
Role	The architecture (the "machine").	The way we train a deep version of that machine.

Remember: Deep = many hidden layers. Shallow NN: 1–2 hidden layers. Deep NN: many hidden layers, huge data, strong computers. Simple features first, complex meaning...

ڈیپ لرننگ مشین لرننگ کا وہ طریقہ ہے جس میں کئی پوشیدہ تہوں والے نیورل نیٹ ورک استعمال ہوتے ہیں۔

5.1.4 Components of Neural Networks ★ Golden

The components of a neural network can be understood through its layered architecture.

- Data enters, gets processed, and exits, with mathematical adjustments happening inside.

#	Component	Description
1	Input layer	Receives the initial raw data (for example image pixels, sound waves, text).
2	Neurons / nodes	Small processing units that receive, calculate, and pass information forward.
3	Hidden layer(s)	The layers between input and output. They process the data and learn patterns from it.
4	Weights	Numbers that show the importance or strength of connections between neurons.
5	Bias	An extra constant value added to help the model adjust its result and improve accuracy.
6	Activation function	A mathematical gate that decides whether the information is important enough to move to the next layer.
7	Output layer	Gives the final computed result, decision or prediction (for example "This is a cat").
8	Loss function	Checks the error (how much the model's answer is wrong compared with reality) so the network can correct itself.

```
x1, x2 = 0.8, 0.5
w1, w2 = 0.4, 0.6
b = 0.1
s = x1 * w1 + x2 * w2 + b
y = 1 if s >= 0.5 else 0
print(round(s, 2))
print(y)
```

Remember: Weights = importance. Bias = extra nudge. Activation = yes/no gate. Loss = how wrong. A large weight means that input matters more. Bias shifts the sum up or...

Exam: Golden questions: "List the components of a neural network" (often 8 names + one line each), "What are weights and bias?", and a numerical neuron: given x , w , b and a ...

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوشیدہ تہ، وزن (Weights)، بائیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔

5.1.5 Components of Deep Learning

Deep learning relies on specific advanced components to achieve its high accuracy.

- These components differentiate it from basic machine learning.

#	Component	What it means
1	Large input data	Deep learning is “ data hungry .” It needs massive datasets (thousands or millions of records) for better learning and accuracy.
2	Many hidden layers	Uses dozens or hundreds of hidden layers to learn deeper, more intricate patterns.
3	Automatic feature learning	Unlike older models where humans had to manually highlight important features, deep learning extracts important features by itself .
4	Learning algorithm	Math routines (like backpropagation) that help the system reduce mistakes and improve learning continuously.
5	High computing power	Processing millions of data points across deep layers needs specialised, strong hardware, such as GPUs (Graphics Processing Units).
6	Training process	The model is trained again and again in loops called epochs to gradually improve its final performance.

Remember: D-L-A-L-G-T. Data (lots), Layers (many), Automatic features, Learning algorithm, GPU power, Training epochs. “Deep Learners Always Love Good Training.”

ڈیپ لرننگ کو عام مشین لرننگ سے یہ چیزیں الگ کرتی ہیں: بہت بڑا ڈیٹا، بہت سی پوشیدہ تہیں، خود بخود خصوصیات نکالنا، سیکھنے کا الگورتھم...

Future of AI, NN and DL in Pakistan

Machine learning, neural networks and deep learning will play a key role in Pakistan's future.

- The National AI Policy 2025 highlights how artificial intelligence is now part of national planning, focusing on innovation,...

Sector	Potential impact in Pakistan
Education	Smart learning systems, online tutoring, automatic assessments, and finding where students need specific help.
Healthcare	Disease detection, medical image analysis, and managing hospital data. Crucial for rural areas lacking specialist doctors.
Agriculture	Monitoring crops, predicting weather, managing irrigation, and estimating yields — highly important for Pakistan's economy.
Business and banking	Fraud detection, customer support (chatbots), market forecasting, and intelligent risk analysis.
Local languages	Speech recognition, translation and text-to-speech tools for Urdu, Sindhi and other regional languages.

Remember: Five sectors: E-H-A-B-L. Education, Healthcare, Agriculture, Business/banking, Local languages. Policy words: innovation, skills, ethics, digital growth.

قومی اے آئی پالیسی 2025 کے مطابق مصنوعی ذہانت پاکستان کی منصوبہ بندی کا حصہ ہے: جدت، مہارت، اخلاقی استعمال اور ڈیجیٹل ترقی۔

5.2 Applications of Neural Networks and Deep Learning

Neural networks and deep learning are widely used in modern life.

- Many intelligent systems use them in the background to recognise information, make predictions and support users.

#	Application	What it does	Example
1	Image recognition	Identifies objects, faces or patterns in images. Used in security cameras, photo search and biometric locks.	Unlocking a smartphone using face recognition; a camera at a Karachi plaza matching a face with a staff list.
2	Speech recognition	Converts spoken words into digital text or commands. Deep learning helps with accents, pronunciation and voice patterns.	A student speaks to Google Assistant or Alexa to set a morning alarm; Urdu voice search on a phone.
3	Natural Language Processing (NLP)	Helps computers understand, process and respond to human language grammatically and in context. Core of chatbots, translators and AI writers.	Using Google Translate to turn a difficult English paragraph into Urdu; a bank chatbot answering in Urdu.
4	Healthcare systems	Analyse complex medical images (X-rays, CT scans, MRIs) to detect diseases early and accurately.	A hospital system highlighting abnormal patterns in a lung X-ray that might indicate pneumonia or tumours — useful at a DHQ hospital that has few radiologists.
5	Recommendation systems	Analyse past behaviour and preferences to suggest content, products or services.	A student watches computer-science videos on YouTube; the platform recommends more programming tutorials.
6	Autonomous vehicles	Self-driving cars use neural networks to process real-time camera and radar data: roads, signals, pedestrians, obstacles.	A self-driving car detects a pedestrian crossing and automatically applies the brakes.
7	Fraud detection	Banks compare current activity with previous spending patterns to catch unusual transactions.	An HBL or Meezan app flags a sudden Rs 80,000 payment from a new city at 3 a.m.
8	Assistive technologies	AI supports people with disabilities.	Speech-to-text tools for users who have difficulty typing; screen readers that speak Urdu text aloud.

Remember: Eight applications: I-S-N-H-R-A-F-A. Image, Speech, NLP, Healthcare, Recommendations, Autonomous vehicles, Fraud, Assistive tech. "I See New Hospitals..."

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیسنگ (NLP)، صحت کے نظام...

5.3.1 Secure Collaboration (Authentication, MFA, Access Control) ★ Golden

Digital tools allow people to share files, attend online classes, edit documents and work together from different locations.

- Collaboration is useful only when data is protected and communication remains safe.
- Secure collaboration means using digital tools to work together in a safe, controlled and responsible way.
- It depends on several key elements.

Remember: A-A-A-P-S. Authentication, Access control, Accountability, Privacy, Safe transmission. MFA = at least two different factor types (know / have / are). 2FA is...

Exam: Typical items: "Define secure collaboration", "Differentiate authentication and access control", "What is MFA/2FA?", "Explain five elements of secure collaboration with..."

محفوظ تعاون کا مطلب ہے ڈیجیٹل ٹولز سے محفوظ، کنٹرول شدہ اور ذمہ دارانہ انداز میں ساتھ کام کرنا۔

5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

Data protection means keeping digital information safe from unauthorised access, misuse, loss, damage or accidental deletion.

3 copies → 2 media types → 1 copy off-site

Plain	S	I	N	D	H
+3	V	L	Q	G	K

Remember: Cipher = lock the message. Password = lock the door. 3-2-1 = spare keys in three places. Firewall = gate. Antivirus = indoor search. Caesar: shift letters;...

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔

5.4.1 Common Security Threats (Malware, Phishing, DoS) ★ Golden

A security threat is any action, event or weakness that can harm a computer system or its data.

- Mitigation techniques are protective measures used to prevent threats or reduce their impact.
- This section names the common threats; later sections show how to spot them and how to reduce the damage.

#	Threat	What it is	Example
1	Malware	Harmful software designed to damage a computer, steal data, or gain unauthorised access. Includes viruses, worms, spyware, ransomware and Trojans .	Installing cracked software that secretly corrupts files, or a "free game" USB that encrypts a college lab's coursework (ransomware).
2	Phishing	A social-engineering trick that deceives users into sharing personal information (passwords, bank details, OTPs) using fake emails or websites that look genuine.	An SMS that looks like JazzCash or HBL asking you to "verify" by typing your OTP on a fake page.
3	Unauthorised access	Entering a system, account or file without permission. Often caused by weak passwords, shared passwords, or leaving devices unlocked.	A classmate using an unlocked lab PC to open someone else's Google Classroom.
4	Data theft / identity theft	Data theft is stealing or copying data without permission. Identity theft is using another person's stolen details (name, CNIC, bank details) to pretend to be them online and commit fraud.	A leaked list of CNICs used to open a SIM or a wallet in someone else's name.
5	Denial of Service (DoS)	A website or system is flooded with thousands of fake requests, causing it to slow down or crash, so real users cannot access the service.	Hackers flooding an online exam portal so students cannot log in on BISE result day or during an e-test.
6	Human error	Mistakes by legitimate users: accidentally deleting files, sending confidential data to the wrong email, or falling for a simple scam.	A teacher emailing the whole marksheet to the wrong parent; a student forwarding an OTP "to help the bank."

Remember: Malware = bad program. Phishing = fake message. DoS = flood. Identity theft = stolen 'you'. Human error = honest mistake. Cracked software and unknown USBs are...

Exam: Expect "Define a security threat", "Explain malware / phishing / DoS with examples", and "Differentiate data theft and identity theft." List all six threats if the...

سیکیورٹی خطرہ وہ عمل یا کمزوری ہے جو نظام یا ڈیٹا کو نقصان پہنچاے۔

5.4.2 Identifying Security Threats

Identifying security threats means noticing signs that data or systems may be at risk.

- Early identification helps you act before serious damage occurs.

Sign / indicator	What it likely means
Slow performance, missing files, or random pop-ups.	Malware or background unauthorised activity.
Failed login attempts or strange login locations.	Attempted unauthorised access .
Emails urging quick action, asking for an OTP, or containing bad grammar / strange links.	Classic phishing .
Unexpected deletion, copying or movement of records.	Data theft or insider misuse.
Sudden massive spike in server requests or a very slow website.	An ongoing DoS attack.

Remember: Slow/pop-ups → malware. Odd logins → unauthorised access. Urgent OTP email → phishing. Files moving → theft. Traffic spike → DoS.

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔

5.4.3 Apply Mitigation Techniques

Mitigation techniques are actions taken to prevent security threats or reduce their harmful effects once identified.

- Technology helps, but it cannot stop all threats if the user is careless.

#	Technique	How it helps
1	Antivirus and firewalls	Antivirus scans and removes harmful software (for example from USB drives). A firewall monitors network traffic and blocks suspicious access from the internet.
2	Strong passwords and 2FA	Complex passwords slow brute-force guessing. 2FA adds an extra layer: an OTP on your phone as well as the password.
3	Data encryption	Converts data into unreadable code during transmission. Protects passwords and sensitive data if they are intercepted.
4	Regular backups	Copies on cloud or external drives allow quick recovery after a crash, human error or ransomware.
5	User awareness (the human factor)	Training users to avoid unknown links, suspicious attachments and password sharing. The human is often the weakest link — and the strongest defence when trained.

Remember: Five shields: A-F, P-2, E, B, U. Antivirus+Firewall, Passwords+2FA, Encryption, Backups, User awareness.

Ransomware is why backups matter: if files are locked,...

تحفظ کی تدابیر خطرات کو روکتی یا ان کا نقصان کم کرتی ہیں: اینٹی وائرس اور فائر وال، مضبوط پاس ورڈ اور دو عنصری تصدیق (2FA)، انکرپشن...

5.5 Equity and Equal Access in Digital Collaboration

Collaboration tools are vital for online learning and teamwork.

- To use these tools fairly, all users must participate without barriers.
- You must understand the difference between equal access and equity.

Remember: Equal = same tools. Equity = extra support for the same outcome. Same shoes for all vs a ramp for the wheelchair user.

مساوی رسائی (Equal Access) سب کو ایک جیسے اوزار دیتی ہے، جیسے ایک ہی زوم لاگ ان۔

5.5.1 Functions of Equity and Equal Access ★ Golden

Equity and equal access are not slogans. They do work in a digital classroom.

- They do work in a digital classroom.
- The source lists four functions.

Function	What it does	Pakistani example
1. Providing need-based support (equity)	Help tailored to individuals.	Audio descriptions for visually impaired students; a screen reader so a blind student in Karachi can hear a digital textbook.
2. Removing barriers	Help students in remote areas attend online classes.	Mobile-friendly, low-data apps so a student in Tharparkar or the katcha areas of Sindh can join without a fibre connection.
3. Enabling active participation	Students who miss live class can still take part.	Recorded lectures so load-shedding or an internet outage in Hyderabad does not make a student fall behind.
4. Supporting different learning needs	Offer more than one format.	Videos, text notes and audio of the same Chapter 5 topic so each student learns in the way that suits them best.

Remember: Four functions: Support, Remove, Participate, Formats. "Some Rural Pupils Fail" without them — Support (need-based), Remove barriers, Participate via...

Exam: Golden question: "Explain the functions of equity and equal access in digital collaboration" (5–8 marks). Write the four headings in order, one example each, and add the...

انصاف اور مساوی رسائی کے چار کام ہیں: ضرورت کے مطابق مدد (جیسے نابینا طالب علم کے لیے آڈیو وضاحت)، رکاوٹیں ہٹانا (دور دراز علاقوں...

5.5.2 Collaboration Tools

Digital collaboration tools are online platforms that support communication, file sharing, learning and teamwork across different locations.

Tool	Primary function and educational use
Google Docs	Write, edit, comment and share documents in real time. Excellent for group assignments.
Zoom	Online meetings, virtual classes, screen sharing and video-based communication.
Microsoft Teams	A unified workspace for online meetings, file sharing, class groups and structured teamwork.
Google Classroom / LMS	Learning Management Systems help teachers manage lectures, distribute assignments, conduct quizzes and grade student activities.
WhatsApp	Quick class announcements, urgent reminders and instant group communication. Must be used responsibly.
Email	Formal communication, official notices, submitting large attachments, and keeping a professional written record.

Remember: Docs = write together. Zoom/Teams = meet. Classroom = official class hub. WhatsApp = fast but informal. Email = formal record.

ڈیجیٹل تعاون کے اوزار مختلف مقامات سے بات چیت، فائل شیئرنگ اور ٹیم ورک چلاتے ہیں: گوگل ڈاکس مشترکہ تحریر کے لیے، زوم اور مائیکروسا...

Career Opportunities

This unit covers neural networks, deep learning, secure collaboration, data protection and digital equity.

- Mastery of these ideas opens high-demand careers in Artificial Intelligence and Cybersecurity — including in Pakistan's banks, software...

Career	What the person does
Machine Learning Engineer	Builds and trains machine-learning and neural-network models used in face recognition, NLP and recommendation systems.
Cybersecurity Analyst	Monitors systems for threats such as malware and phishing, and applies mitigation (firewalls, 2FA) to protect organisational data.
AI Researcher	Studies and develops new techniques in artificial intelligence and deep learning, often in university labs or high-tech R&D centres.
Data Privacy Officer	Ensures the organisation's data handling complies with legal privacy principles, protecting sensitive user information and secure collaboration.
Ethical Hacker	Legally tests systems and networks for vulnerabilities (penetration testing) before malicious attackers can exploit them via DoS or malware.

Remember: AI careers lean on Python, maths and ethics. Cyber careers lean on threats, 2FA, firewalls and careful testing. Both need clear communication.

اس باب سے جڑے پیشے ہیں: مشین لرننگ انجینئر، سائبر سیکیورٹی اینالسٹ، اے آئی محقق، ڈیٹا پرائیویسی آفیسر اور اخلاقی ہیکر۔

Key terms

Term	Meaning	اردو
Artificial intelligence (AI)	The broad field in which machines perform tasks that normally need human thinking.	مصنوعی ذہانت
Machine learning (ML)	A model that learns patterns from data and then predicts on new data.	مشین لرننگ
Neural network (NN)	A brain-inspired mathematical model of connected neurons in layers.	نیورل نیٹ ورک
Deep learning (DL)	An ML method that trains neural networks with many hidden layers.	ڈیپ لرننگ
Weight	A number showing the strength or importance of a connection between neurons.	وزن (وزنِ تعلق)
Bias	An extra constant added to a neuron's sum to improve accuracy.	بائیس
Activation function	A gate that decides whether a signal is important enough to pass on.	ایکٹیویشن فنکشن
Loss function	A measure of how wrong the model's answer is compared with reality.	لاس فنکشن
Epoch	One full training pass through the dataset.	ایپوک (تربیتی چکر)
GPU	Graphics Processing Unit — specialised hardware used to train deep networks.	جی پی یو
NLP	Natural Language Processing — computers understanding and producing human language.	قدرتی زبان کی پروسیسنگ
Authentication	Confirming a user's identity before access (password, PIN, OTP, biometrics).	تصدیقِ شناخت
MFA / 2FA	Multi-/two-factor authentication: two or more different proofs of identity.	کثیر / دو عنصری تصدیق
Access control	Rules for what an authenticated user may do (view, edit, share, ...).	رسائی پر قابو
Encryption	Turning plaintext into ciphertext so unauthorised people cannot read it.	رمز کاری (انکریپشن)
Caesar cipher	A symmetric cipher that shifts letters by a fixed number of places.	سیزر سائفر

Term	Meaning	اردو
3-2-1 backup	Three copies, two media types, one copy stored off-site.	3-2-1 بیک اپ
Firewall	A barrier that monitors network traffic and blocks suspicious access.	فائر وال
Malware	Harmful software such as viruses, worms, spyware, ransomware and Trojans.	میلویئر
Phishing	Fake messages or sites that trick people into giving passwords or OTPs.	فشنگ
Denial of Service (DoS)	Flooding a service with fake requests so real users cannot use it.	سروس سے انکار
Equal access	Giving everyone the same tools and opportunity.	مساوی رسائی
Equity	Giving extra, specific support so people facing barriers reach the same outcome.	انصاف (ایکویٹی)
LMS	Learning Management System — platform for lectures, assignments, quizzes and grades.	تعلیمی نظم کا نظام

Chapter in one look

- **ML** is a model (a trained brain) that learns from data in three steps: input, learning, prediction on new unseen data — without a hand-written rule for every case.
- **AI $\supset$ ML $\supset$ DL**: deep learning sits inside machine learning, which sits inside artificial intelligence.
- A **neural network** is a brain-inspired mathematical model of connected neurons in an input layer, hidden layer(s) and output layer (face unlock: camera $\rightarrow$ features $\rightarrow$ unlock).
- **Deep learning** trains neural networks with many hidden layers; early layers learn simple features, deeper layers learn complex patterns. It needs huge data and computing power.
- NN components: input layer, neurons, hidden layers, weights, bias, activation function, output layer, loss function. A neuron computes $s = \sum(x \times w) + b$, then an activation gate.
- DL components: large data, many hidden layers, automatic feature learning, learning algorithm (backpropagation), GPUs, training epochs. Pakistan's National AI Policy 2025 covers education, health, agriculture, banking and local languages.
- Applications: image and speech recognition, NLP, healthcare imaging, recommendations, autonomous vehicles, fraud detection, assistive tech.
- **Secure collaboration**: authentication (who), MFA/2FA (two or more factors), access control (what you may do), accountability, privacy, safe transmission.
- **Data protection**: encryption (plaintext $\rightarrow$ ciphertext; symmetric vs asymmetric; Caesar shift), strong passwords, 3-2-1 backup, firewall and antivirus.
- Threats: malware, phishing, unauthorised access, data/identity theft, DoS, human error. Spot them by pop-ups, odd logins, urgent OTP emails, missing files, traffic spikes.
- Mitigation: antivirus + firewall, strong passwords + 2FA, encryption, regular backups, user awareness.
- **Equal access** = same tools for all; **equity** = extra support for the same outcome. Functions: need-based support, removing barriers, recorded participation, multiple formats. Tools: Docs, Zoom, Teams, Classroom/LMS, WhatsApp, email.

Entrepreneurship in the Digital Age

★ 5 Golden · 18 cards · Entrepreneurship in the Digital Age · Prototype · Prototype Development Cycle · Minimum Viable Product (MVP) · Beachhead Market

6.1 Entrepreneur and Entrepreneurship

An entrepreneur is a person who identifies a problem, thinks of a solution, and tries to change that solution into a product or service.

1. Problem → 2. Solution → 3. Product

- They observe people and listen to their needs. They notice what annoys people again and again.
- They take calculated risks. A calculated risk is a risk you take after thinking about what you could lose and how to keep that loss small....
- They improve their idea step by step. They do not expect the first attempt to be perfect.

Remember: Simple meaning. An entrepreneur asks: "What problem can I solve?" Entrepreneurship is the process of testing and improving that solution until it becomes...

کاروباری شخص (Entrepreneur) وہ ہوتا ہے جو کسی مسئلے کو پہچانتا ہے، اس کا حل سوچتا ہے اور اس حل کو کسی پراڈکٹ یا سروس میں بدلنے کی کوشش کرتا ہے۔

6.2 Entrepreneurship in the Digital Age ★ Golden

Entrepreneurship in the digital age means using digital tools to start, test, improve, and grow a business idea.

- These tools may include mobile phones, social media, websites, online forms, digital payments, maps, videos, and simple apps.
- In Pakistan you already know many of them: WhatsApp, Facebook, Instagram, TikTok, YouTube, Google Forms, Google Maps, and digital payments...

Remember: Remember "S-T-I-G": digital tools help you Start, Test, Improve and Grow. Think: "Start The Idea Growing."

Exam: Common questions: "Define entrepreneurship in the digital age", "How have digital tools changed entrepreneurship?" and "Give examples of digital entrepreneurship without..."

ڈیجیٹل دور میں انٹرپرائیورسپ کا مطلب ہے موبائل فون، سوشل میڈیا، ویب سائٹ، آن لائن فارم اور ڈیجیٹل ادائیگی جیسے ٹولز کے ذریعے کاروبار...

Local Examples of Digital Entrepreneurship (Table 6.1)

Digital entrepreneurship can begin from ordinary problems.

- The examples below are close to the context of Sindh and Pakistan, where many students, parents, and small service providers already use...

Problem	Digital Idea	Possible First Users	Urdu Context
Parents miss college notices.	WhatsApp notice and reminder service.	Parents of first-year students.	والدین تک کالج کے نوٹس پہنچانے کے لیے واٹس ایپ سروس۔
Small embroidery shop has few customers.	Simple catalogue with order form on Facebook or WhatsApp.	Customers in the same city.	کڑھائی کی دکان کے لیے واٹس ایپ کیٹلاگ۔
Students forget test dates.	Google Calendar guide or simple reminder app.	Grade XII students.	ٹیسٹ کی تاریخیں یاد دلانے والی ایپ یا کیلنڈر۔
Canteen has long queues during break.	Pre-order form for snacks.	Students and teachers.	کینٹین کے لیے پری آرڈر فارم۔

Remember: Think "Problem first, tool second." Start with the pain, not with the technology.

جدول 6.1 بتاتی ہے کہ ڈیجیٹل کاروبار عام مسائل سے شروع ہو سکتا ہے، جیسے والدین کا کالج نوٹس نہ دیکھ پانا، کڑھائی کی دکان پر کم گاہک...

6.3 From Problem to Business Idea

A good business idea usually starts from observing daily life.

Problem → Idea → Prototype → Feedback → MVP

- What problem do people face?
- Who faces this problem?
- How often does it happen?
- What do people currently do to solve it?

Step	What Students Do	Example
Observe	Notice repeated problems around home, college, or market.	Students forget practical file dates.
Ask Users	Talk to people who face the problem.	Ask five classmates how they remember deadlines.
Select One Problem	Choose one clear problem only.	Practical file reminder.
Suggest Solution	Think of a simple digital or paper-based solution.	Reminder form or app screen.
Test Early	Show the idea to users before final development.	Ask classmates if they will use it.

Remember: Mnemonic for Table 6.2: "Only Ask Simple Solutions Today": Observe, Ask users, Select one problem, Suggest solution, Test early.

اچھا کاروباری خیال روزمرہ زندگی کے مشاہدے سے شروع ہوتا ہے۔

6.4 Prototype ★ Golden

A prototype is an early model of a product, service, website, or app.

- It shows how the idea may look or work.
- It is NOT the final product.
- It is made to test the idea and collect feedback.

Remember: Prototype = "practice model". It is like a rough draft of an essay: you show it, get corrections, and rewrite it before the final copy.

Exam: Expect "Define prototype and write its benefits" (3–5 marks). Write: (1) definition with the words early model, not the final product, test the idea and collect...

پروٹوٹائپ (Prototype) کسی پراڈکٹ، سروس، ویب سائٹ یا ایپ کا ابتدائی نمونہ ہوتا ہے جو دکھاتا ہے کہ خیال کیسا نظر آئے گا یا کیسے کام کرے گا۔

6.5 Types of Prototypes

Prototypes are commonly divided into three types according to their level of detail, called fidelity.

Low-Fi (Paper) → Mid-Fi (PowerPoint) → High-Fi (Figma/HTML)

- Fidelity means how close the prototype is to the final product.
- Low fidelity = far from final; high fidelity = very close to final.

Type	Meaning	Best Use	Example
Low-Fidelity	A quick and rough model of the idea.	Very early testing.	Hand-drawn app screens for homework reminder.
Mid-Fidelity	A clearer model that shows layout and flow.	Testing navigation.	Clickable screens in PowerPoint or Figma.
High-Fidelity	A detailed model close to the final product.	Testing realistic experience.	Working demo of a small order form website.

Remember: Fidelity = "faithfulness" to the final product. Remember Paper → PowerPoint → Polished (low → mid → high).

فڈیلیٹی (Fidelity) کا مطلب ہے کہ پروٹوٹائپ حتمی پراڈکٹ سے کتنا ملتا جلتا ہے۔

6.6 Prototype Development Cycle ★ Golden

Prototype development is not a one-time activity.

- A prototype is created, tested, improved, and tested again.
- This repeating process is called iteration.
- The main stages are Design, Build, Test, and Iterate.

Stage	What happens
1. Design	Understand the problem and plan the solution. Decide what the prototype should show (e.g. screen flow, features).
2. Build	Create the first version (physical or digital). The aim is not beauty; the aim is to make the idea visible.
3. Test	Show the prototype to users. Observe what they understand and what they find difficult.
4. Iterate	Improve the prototype based on user feedback. Remove unnecessary parts and add missing features.

Remember: Mnemonic: "Do Big Things Iteratively" = Design, Build, Test, Iterate. And remember the arrow at the end goes back to Design: it is a cycle, not a line.

Exam: A favourite long question: "Explain the prototype development cycle with a diagram." Draw the four boxes in a circle with arrows, write one or two lines for each stage...

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔

6.7 Class Activity: Create a Prototype

Business idea scenario. Students often forget homework deadlines, test dates, and required class items.

1. Understand → 2. Draw low-fi → 3. Test → 4. Analyse feedback → 5. Improve

- Students often forget homework deadlines, test dates, and required class items.
- A group of students wants to design a simple "Homework Helper" or "ClassBuddy" reminder system.
- The team will not immediately build a complete mobile app.
- They will first understand the problem, draw a low-fidelity prototype, test it with classmates, and improve it.

Finding	Meaning	Design Decision
7 out of 10 forget tasks.	The problem is common.	App should show a daily task list.
Reminders needed before leaving home.	Reminder time is important.	Add morning notification option.
Students want simple apps.	Design must be simple.	Use large buttons and clear labels.

Remember: Feedback → Problem → Improvement. Every comment should be turned into a problem statement and then one clear change, exactly like the three columns of Table...

اس سرگرمی میں طلبہ "کلاس بڈی" (ClassBuddy) نامی یاد دہانی نظام بناتے ہیں۔

6.8 Minimum Viable Product (MVP) ★ Golden

A Minimum Viable Product (MVP) is the simplest working version of a product.

- Minimum: it has only the essential features. No extras.
- Viable: it is usable enough to actually solve the main problem. ("Viable" means able to work successfully.)

Complete canteen app (final product)	Canteen MVP
Login, online payment, delivery tracking, discount offers...	A simple Google Form where students select snacks before break.

Remember: Simple meaning. An MVP answers: "Will people actually use this solution?" It tests real use and behaviour, not only the visual design. Analogy: before opening...

Exam: Typical questions: "What is an MVP? Explain the words Minimum and Viable" and "Give an example of an MVP." Write the full form, the definition (simplest working version,...

ایم وی پی (Minimum Viable Product) کسی پراڈکٹ کا سب سے سادہ مگر کام کرنے والا ورژن ہے جس میں صرف ضروری خصوصیات ہوتی ہیں۔

6.9 Difference between Prototype and MVP

Prototype and MVP are related, but not the same.

- A prototype is mainly used to show and test the design of an idea.
- An MVP is used to test real use and market demand.
- Market demand means whether enough people want the product and will use (or pay for) it.

Point	Prototype	MVP
Main purpose	To show how the idea may look or work.	To test whether users will actually use the solution.
Working status	May not work; can be a sketch or model.	Must work at least in a basic way.
Users	Small group gives feedback on design.	Early real users test the product in daily life.
Cost	Very low cost.	Low cost but may need some basic development.
Example	Paper screens for a canteen order system.	Google Form used for real canteen orders for three days.

Remember: Key takeaway Prototype = "Does it look right and make sense?" MVP = "Will people actually use it and find value in it?"

Memory hook: Prototype → Picture of the...

پروٹوٹائپ اور ایم وی پی آپس میں جڑے ہیں مگر ایک جیسے نہیں۔

6.10 Identifying the Riskiest Assumption

Every business idea is based on assumptions.

- An assumption is something we believe to be true, but have not tested yet.
- The riskiest assumption is the belief that can make the entire business fail if it turns out to be wrong.
- Entrepreneurs should identify and test the riskiest assumption early, because it saves time and money.

Business Idea	Possible Risky Assumption	How to Test It (MVP)
Online tuition finder	Students will pay for online tutoring.	Ask 20 students and offer a trial class to see if they sign up.
Canteen pre-order form	Students will order snacks before break.	Test a Google Form for one week in real time.
Ajrak online catalogue	Customers will place orders through WhatsApp.	Share catalogue with 30 customers and record actual orders.
College van update system	Parents and students will check van updates daily.	Create a trial WhatsApp broadcast group for one route.

Remember: Ask: "What would kill this idea?" The answer is usually the riskiest assumption. Test it first, cheaply, with real behaviour.

ہر کاروباری خیال کچھ مفروضوں (Assumptions) پر قائم ہوتا ہے، یعنی وہ باتیں جنہیں ہم سچ سمجھتے ہیں مگر ابھی آزمایا نہیں۔

6.11 Developing an MVP

To develop an MVP, the team must select only the essential features.

- Extra features should be avoided at the beginning.
- A common mistake is trying to build everything at once.
- This makes the product difficult, costly, and slow.

MVP Element	Question to Ask	Example Answer
Problem	What problem are we solving?	Students wait too long at canteen.
Target users	Who will use it first?	Grade XII students in one college.
Main feature	What is the ONE feature that solves it?	Pre-order snacks before break.
Basic design	What steps are needed?	Menu, name, class, order time.
Cost	What free/low-cost tools to use?	Google Form or printed slips.
Marketing	How will early users hear about it?	WhatsApp message to class group.
Feedback	How will users respond?	Short feedback form after 3 days.

Remember: Mnemonic for the 8 elements: "People Try My Bakery; Cheap Muffins Find Improvement": Problem, Target users, Main feature, Basic design, Cost, Marketing,...

ایم وی پی بناتے وقت صرف ضروری خصوصیات چنیں؛ سب کچھ ایک ساتھ بنانے سے کام مشکل، مہنگا اور سست ہو جاتا ہے۔

6.12 Case Study: MVP for a Canteen Pre-Order System

Problem: In a college, students wait in long lines at the canteen.

Paper prototype → Feedback: short menu, pay at collection → Google Form MVP → 3-day test → Improve

- Sometimes snacks finish before students reach the counter.
- A group wants to test a canteen pre-order idea.
- Goal: Their aim is NOT to create a complete app immediately.

MVP Element	Decision
Features	Menu list, student name, class, item, and quantity.
Cost & Tools	Free Google Form or printed order slips.
Marketing	Class announcement and WhatsApp group message.
Testing period	Three days.
Feedback method	Ask students if they saved time and would use it again.
Improvement	Keep only 5 items; close orders 20 minutes before break.

Remember: Remember the case study as "5 items, 3 days, 20 minutes": five menu items, a three-day test, orders closing twenty minutes before break.

ایک کالج میں طلبہ کینٹین کی لمبی قطاروں میں انتظار کرتے تھے۔

6.13 Testing an MVP

MVP testing means giving the basic working version to real users and studying their response.

- Numbers (quantitative data) show how many users tried the MVP and how many may use it again.
- Comments (qualitative data) explain what users liked, disliked, or found confusing.
- How many users will test it.
- How long the test will continue.

Type of Feedback	Question	Example Result
Number-based	How many users tried it?	40 students used the form.
Number-based	How many would use it again?	30 out of 40 said yes.
Comment-based	What was confusing?	Did not know order closing time.
Observation	Where did users face difficulty?	Forgot to enter class name.

Remember: Data skill link. Feedback is useful only when it is studied: Count the responses → Look for repeated problems → Decide which improvement comes first. Remember...

ایم وی پی کی جانچ کا مطلب ہے کہ بنیادی کام کرنے والا ورژن اصل صارفین کو دیا جائے اور ان کے ردعمل کا مطالعہ کیا جائے۔

6.14 Beachhead Market ★ Golden

A beachhead market is a small and focused group of first users.

- These users have a clear need and are likely to try the product early.
- Instead of trying to serve everyone, entrepreneurs first focus on one small group.
- This helps them test the MVP properly and improve it before expanding.

Business Idea	Weak First Market	Better Beachhead Market
Study app	All students in Pakistan.	Grade XII students preparing for exams in one college.
Canteen pre-order	All canteens in the city.	One college canteen during lunch break.
Ajrak catalogue	All online shoppers.	Existing customers of one local shop.
Van update system	All transport users.	Students and parents of ONE specific college van route.

Remember: Remember S-R-U: Specific, Reachable, Urgent need. If you cannot meet your first users within a week, your market is too big.

Exam: Asked as "Define beachhead market and give its characteristics" or "Suggest a beachhead market for ...". Write the definition (small, focused group of first users with a...

بیچ ہیڈ مارکیٹ (Beachhead Market) پہلے صارفین کا ایک چھوٹا اور مخصوص گروہ ہے جسے حل کی واضح اور فوری ضرورت ہو اور جو پراڈکٹ جلد آزمانے پر تیار ہو۔

6.15 Complete Student Project

You will work in groups and complete one small entrepreneurship project.

- The project should be simple, local, and possible in the classroom.
- The purpose is to practise the complete journey: problem → prototype → feedback → MVP plan → risky assumption → beachhead market.
- Suggested project ideas: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp...

Criteria	Excellent	Satisfactory	Needs Improvement
Problem ID	Clear, local, and important.	Understandable but needs detail.	Vague or too broad.
Prototype	Shows clear flow and features.	Shows the basic idea.	Incomplete or unclear.
Feedback & Fixes	Collected properly; changes clearly based on feedback.	Collected but analysis is weak; some changes made.	No clear feedback collected; changes not explained.
MVP Plan	Essential features and clear testing plan.	Basic features but needs testing clarity.	Confused with the final product.

Remember: Use the rubric before you submit: read your project sheet row by row and ask, "Is this Excellent yet?" For "Feedback & Fixes", always write which feedback...

طلبہ گروپوں میں ایک چھوٹا، مقامی اور آسان منصوبہ مکمل کرتے ہیں جس میں مسئلے سے پروٹوٹائپ، رائے، ایم وی پی منصوبہ، سب سے خطرناک مفر...

6.16 Ethical and Safe Use of Digital Tools

Digital entrepreneurship should be responsible.

- Privacy: people's personal information belongs to them.
- Truthfulness: report results honestly; do not make up numbers or hide bad feedback.
- Fairness: treat all users equally.

Remember: Ask yourself "P-T-F-E": Is it Private, Truthful, Fair, and Equal-access?

ڈیجیٹل کاروبار ذمہ داری سے کرنا چاہیے۔

Career Opportunities

This unit covered entrepreneurship in the digital age, turning problems into business ideas, prototyping, building MVPs, and identifying beachhead markets.

- Mastery of these concepts opens the door to careers in startups and digital innovation.
- Pakistan has many incubators (for example, National Incubation Centres, NICs, in major cities) that support young founders.

Career	What the person does
Tech Entrepreneur / Startup Founder	Identifies a real-world problem and builds a digital product or service to solve it, guiding it from prototype through to MVP and market launch.
Product Manager	Guides a product's direction by identifying user needs, prioritising features, and testing ideas through prototypes and MVPs.
Business Analyst	Analyses business problems and processes to recommend digital solutions, often testing assumptions before full investment.
Innovation / Startup Consultant	Advises businesses or founders on validating ideas, building MVPs, and identifying beachhead markets.
Digital Marketing Specialist	Plans how a new product or MVP will reach its first (beachhead) market and grow its user base.
Venture / Incubator Analyst	Evaluates early-stage startup ideas and prototypes for potential investment or incubation support.

Remember: Every career in the table uses the same chapter skills: understand users, test assumptions, start small.

اس باب کی مہارتیں اسٹارٹ اپ اور ڈیجیٹل جدت کے شعبوں میں روزگار کے دروازے کھولتی ہیں، جیسے اسٹارٹ اپ بانی، پراڈکٹ مینیجر، بزنس اینا...

Key terms

Term	Meaning	اردو
Entrepreneur	A person who identifies a problem, thinks of a solution, and turns it into a product or service.	کاروباری شخص
Entrepreneurship	The process of planning, starting, testing and improving a business idea.	کاروبار کاری
Calculated risk	A risk taken after carefully thinking about the possible loss and keeping it small.	سوچا سمجھا خطرہ
Digital entrepreneurship	Using digital tools to start, test, improve and grow a business idea.	ڈیجیٹل کاروبار کاری
Business idea	A proposed product or service that solves a real problem for real users.	کاروباری خیال
Target users	The people who will use the product first.	ہدف صارفین
Prototype	An early model of a product, service, website or app made to test the idea and collect feedback.	ابتدائی نمونہ
Fidelity	How close a prototype is to the final product.	اصل سے مطابقت کا درجہ
Low-fidelity prototype	A quick, rough paper model showing the main idea and flow.	کم تفصیل والا نمونہ
Mid-fidelity prototype	A clearer model in PowerPoint, Canva or Figma showing layout and navigation without final graphics.	درمیانی تفصیل والا نمونہ
High-fidelity prototype	A detailed model close to the final product with realistic colours, buttons, data and working features.	زیادہ تفصیل والا نمونہ
Wireframe	A simple outline drawing of a screen showing where items go, without colours or pictures.	اسکرین کا خاکہ
Navigation	How a user moves from one screen to another.	اسکرینوں کے درمیان آمد و رفت
Iteration	Repeating the cycle of creating, testing and improving.	بار بار دہرانا
Feedback	Users' responses and comments about a product or idea.	رائے
Minimum Viable Product (MVP)	The simplest working version of a product with only the essential features, used to test the idea with real users.	کم از کم قابل عمل پراڈکٹ

Term	Meaning	اردو
Viable	Usable enough to actually solve the main problem.	قابلِ عمل
Market demand	Whether enough people want and will use or pay for a product.	بازار کی طلب
Assumption	Something we believe to be true but have not tested yet.	مفروضہ
Riskiest assumption	The belief that would make the whole business fail if it turned out to be wrong.	سب سے خطرناک مفروضہ
MVP plan	A plan stating problem, target users, main feature, design, cost, marketing, feedback and improvement.	ایم وی پی کا منصوبہ
Number-based feedback	Feedback in counts, e.g. how many tried the MVP or would use it again.	عددی رائے
Comment-based feedback	Feedback in words explaining likes, dislikes or confusion.	تبصراتی رائے
Observation	Watching users to see where they face difficulty.	مشاہدہ
Beachhead market	A small, focused group of first users with a clear need who are likely to try the product early.	ابتدائی مخصوص مارکیٹ
Rubric	A marking guide describing levels of quality for each criterion.	جانچ کا معیار نامہ
Privacy	A person's right to control their personal information.	رازداری
Equity	Making sure people with fewer resources or language difficulty can also use a solution.	مساوی رسائی
Startup	A new, small business built around a new product or service idea.	نیا کاروبار

Chapter in one look

- **Entrepreneur:** a person who identifies a problem, thinks of a solution, and turns it into a product or service; observes people, takes calculated risks, improves step by step.
- **Entrepreneurship:** the process of planning, starting, testing and improving a business idea until it becomes useful (Problem → Solution → Product).
- **Digital age:** entrepreneurs use accessible digital tools (social media, forms, websites, WhatsApp, digital payments) to start and test ideas quickly with low investment; often no coding is needed.
- A digital idea must solve a **real problem for real users**, not just sound modern (Table 6.1 examples: notices, embroidery shop, test dates, canteen queues).
- **Business ideas** come from observing daily life and talking directly to users: Observe → Ask Users → Select One Problem → Suggest Solution → Test Early. Start with a small, clear problem.
- **Prototype:** an early model used to show and test the design of an idea; not the final product; makes the idea visible, finds mistakes early, saves time and money.
- **Fidelity** is how close a prototype is to the final product: Low (paper), Mid (PowerPoint/Canva/Figma, tests navigation), High (realistic, Figma/HTML, only after the idea is tested).
- **Development cycle:** Design → Build → Test → Iterate, repeated (iteration). Feedback is crucial; a prototype fails only if useful feedback is ignored.
- **MVP:** the simplest working version with only essential features (Minimum) that still solves the main problem (Viable); it tests real use and market demand.
- **Prototype vs MVP:** prototype asks "Does it look right?"; MVP asks "Will people actually use it?" A prototype may not work; an MVP must.
- **Riskiest assumption:** the untested belief that, if wrong, will make the business fail; test it early with real behaviour.
- **MVP plan:** problem, target users, ONE main feature, basic design, cost, marketing, feedback, improvement plan. Do not build everything at once.
- **Canteen case study:** paper prototype → Google Form MVP with 5 items for 3 days → staff found too many items hard → kept 5 items, separate line, orders close 20 minutes before break.
- **MVP testing:** plan users, duration and data in advance; collect numbers (40 tried, 30 would return = 75%) and comments/observations; count, look for repeated problems, fix the most important first.
- **Beachhead market:** a small, focused, reachable group of first users with a strong, immediate need; win it first, then expand.
- **Ethics and equity:** respect privacy, truthfulness and fairness; collect no unnecessary data; share nothing without permission; design for weak internet, limited devices and Urdu/Sindhi speakers.

NIGHT BEFORE

One-page recaps

Read each recap, close the booklet, say it aloud. Then stop.

Chapter 1 recap — Human Computer Interaction

- HCI = study of how people and computers work together so systems are usable, useful and satisfying.
- Traditional interaction = keyboard, mouse, menus. Natural interaction = touch, voice, gesture, gaze.
- Components: user, computer, interaction (input + output + feedback) inside an environment.
- UI = how it looks and the controls. UX = how the whole experience feels. Wireframe → prototype (Figma) → test.
- Evaluate with usability tests, heuristic evaluation, A/B tests and automated checks. Golden: accessibility, HCI problems, UI vs UX, testing methods.

Chapter 2 recap — Algorithms and Data Structures

- Correctness: trace table (variable values after every step) or stepwise reasoning.
- Clarity: modularity (small named parts) and readability (names, comments, indentation).
- Efficiency: count steps, conditions, repetitions. Big O: $O(1)$ constant, $O(\log n)$ binary search, $O(n)$ linear, $O(n \log n)$ good sorts, $O(n^2)$ nested loops.
- Linear DS: array, linked list, stack (LIFO, push/pop), queue (FIFO, enqueue/dequeue). Non-linear: tree (one root, parent-child), graph (nodes + edges).
- Pick the structure from the problem: undo → stack; printer jobs → queue; hierarchy → tree; maps → graph; fast index → array.

Chapter 3 recap — Python Data Structures, Functions and Files

- list [] ordered, mutable. tuple () ordered, immutable. set {} unordered, unique. dict {k:v} key-value, mutable.
- Functions: def, parameters, return. Local vs global scope. Built-in vs user-defined.
- Files: "r" read, "w" overwrite, "a" append. Always prefer with open(...) as f so the file closes itself.
- Exceptions: try / except / else / finally. Catch specific errors (ValueError, FileNotFoundError).

Chapter 4 recap — Data and Analysis

- SQLite in Python: connect → cursor → execute → commit → close. Always commit writes.
- Pandas DataFrame = table. load CSV/Excel, isnull, dropna, fillna, sort, filter, groupby.
- Charts: bar (categories), line (trend over time), pie (parts of a whole, few slices), histogram (distribution), scatter (relationship of two variables), box (spread and outliers).
- Mean = average. Median = middle after sorting. Mode = most frequent. Range = max-min. Variance / std-dev measure spread. pandas .std() uses sample $n-1$.

Chapter 5 recap — AI, Security and Collaboration

- AI ⊃ Machine Learning ⊃ Deep Learning. A neuron: weighted sum + bias, then an activation function. Deep = many hidden layers.
- Secure collaboration: authentication (passwords, MFA), access control (who may do what).
- Protect data: encryption, strong passwords, 3-2-1 backup, firewall. Threats: malware, phishing, DoS/DDoS, social engineering.
- Equity: everyone can take part (devices, language, disability, bandwidth). Tools: shared docs, video, LMS, versioning.

Chapter 6 recap — Entrepreneurship in the Digital Age

- Entrepreneur = person who turns a problem into a venture, taking risk. Digital tools make testing cheap (WhatsApp, Forms, a Facebook page).
- Prototype = experimental model. Low / mid / high fidelity. Cycle: Design → Build → Test → Iterate.
- MVP = smallest product that tests the riskiest assumption with real users. Prototype ≠ MVP (MVP is used in the real market).
- Beachhead market = the first, narrow group you can win. Measure retention, not only sign-ups. Use data ethically.